US 20250273157A1

(19) United States
(12) Patent Application Publication
KIM et al.
(10) Pub. No.: US 2025/0273157 A1
(43) Pub. Date: Aug. 28, 2025

(54) DISPLAY DEVICE AND METHOD OF DRIVING THE SAME

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(21) Appl. No.: **18/952,360**

(22) Filed: **Nov. 19, 2024**

(30) **Foreign Application Priority Data**

Feb. 27, 2024 (KR) 10-2024-0028278

Publication Classification

(51) **Int. Cl.**
G09G 3/3258 (2016.01)
G09G 5/10 (2006.01)

(52) **U.S. Cl.**
CPC ***G09G 3/3258*** (2013.01); ***G09G 5/10*** (2013.01); *G09G 2300/0842* (2013.01); *G09G 2300/0866* (2013.01); *G09G 2310/0243* (2013.01); *G09G 2310/06* (2013.01)

(57) **ABSTRACT**

A display device includes: a timing controlling unit generating an image data, a data control signal and a gate control signal; a data driving unit generating a data signal using the image data and the data control signal; a gate driving unit generating a plurality of scan signals and an emission signal using the gate control signal; and a display panel displaying an image using the data signal, the plurality of scan signals and the emission signal, wherein the emission signal includes a plurality of cycles each having a duty-on period and an duty-off period during one frame, and wherein duty ratios of at least two of the plurality of cycles are different from each other.

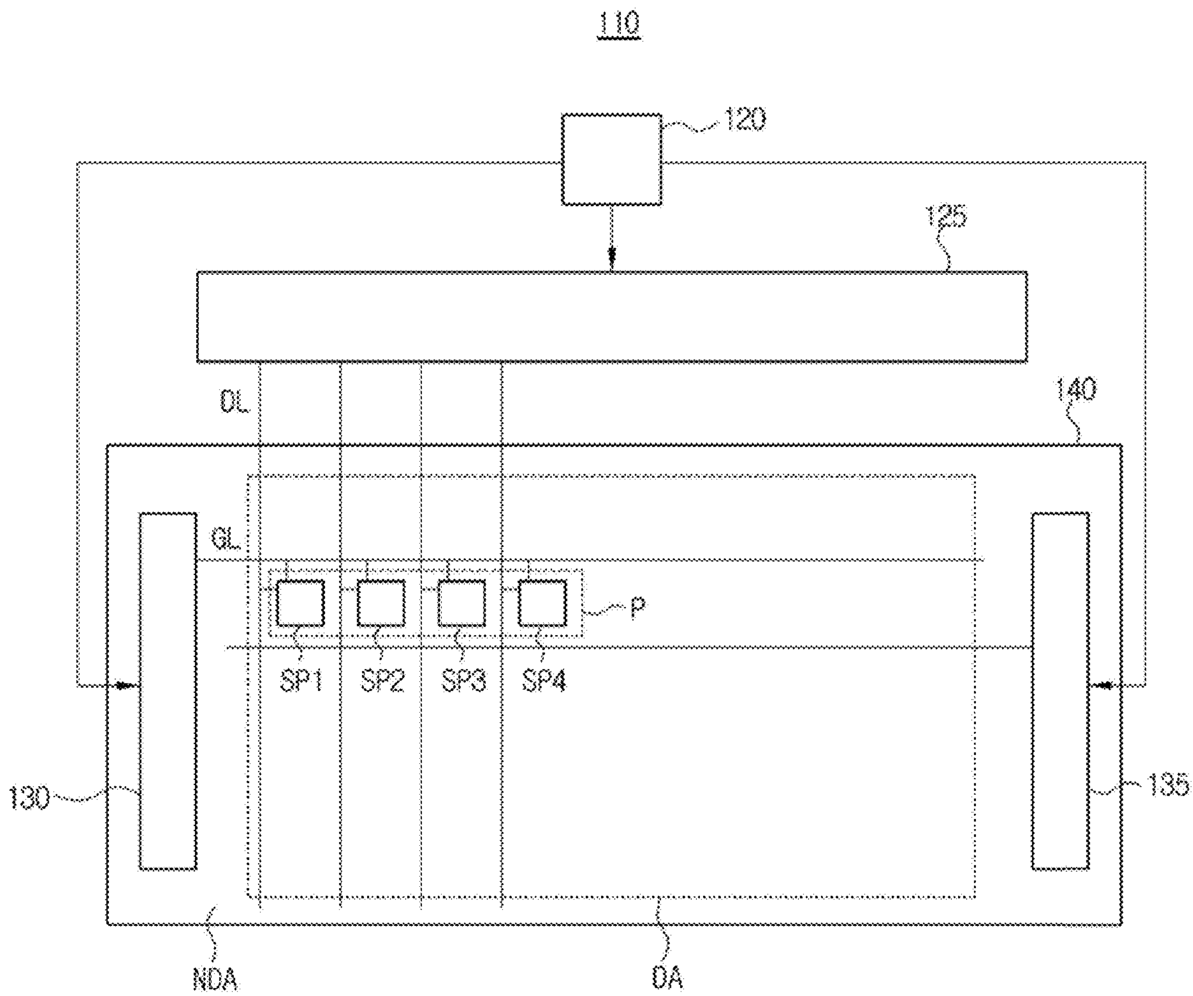


FIG. 1

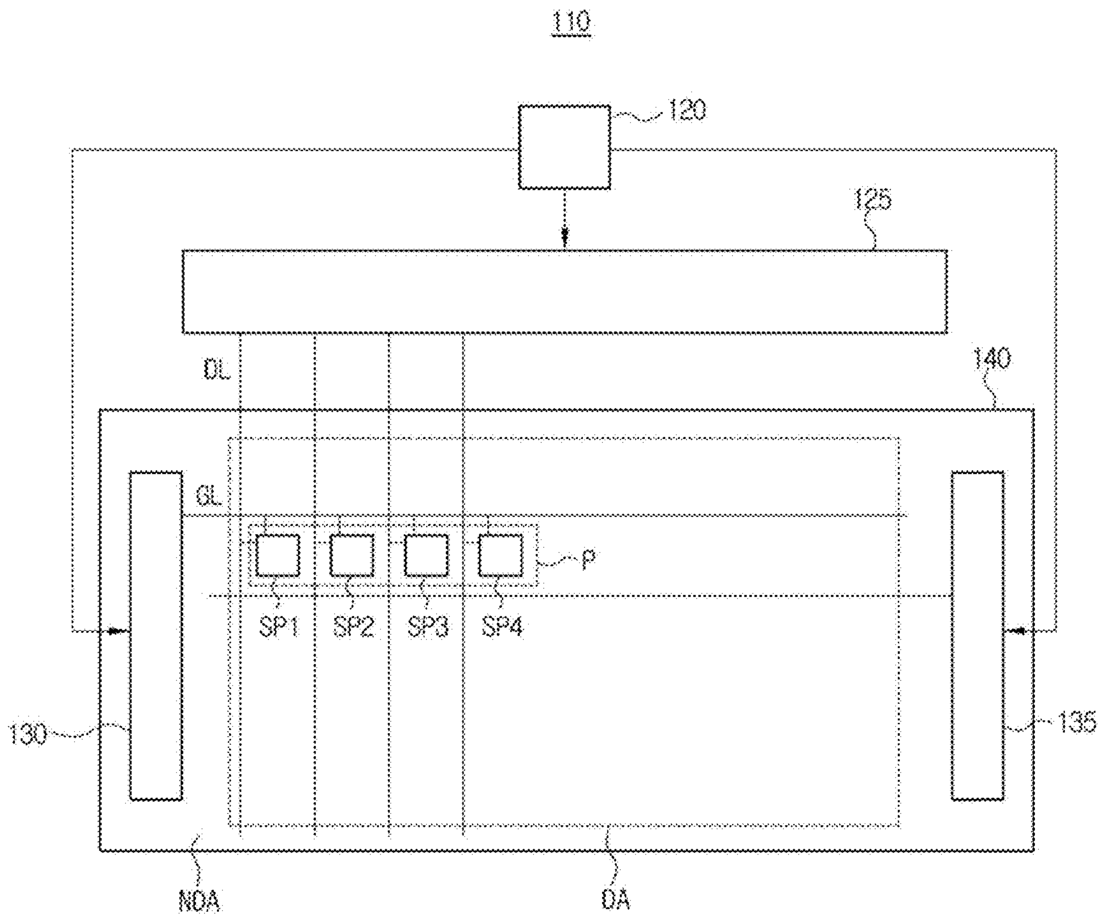
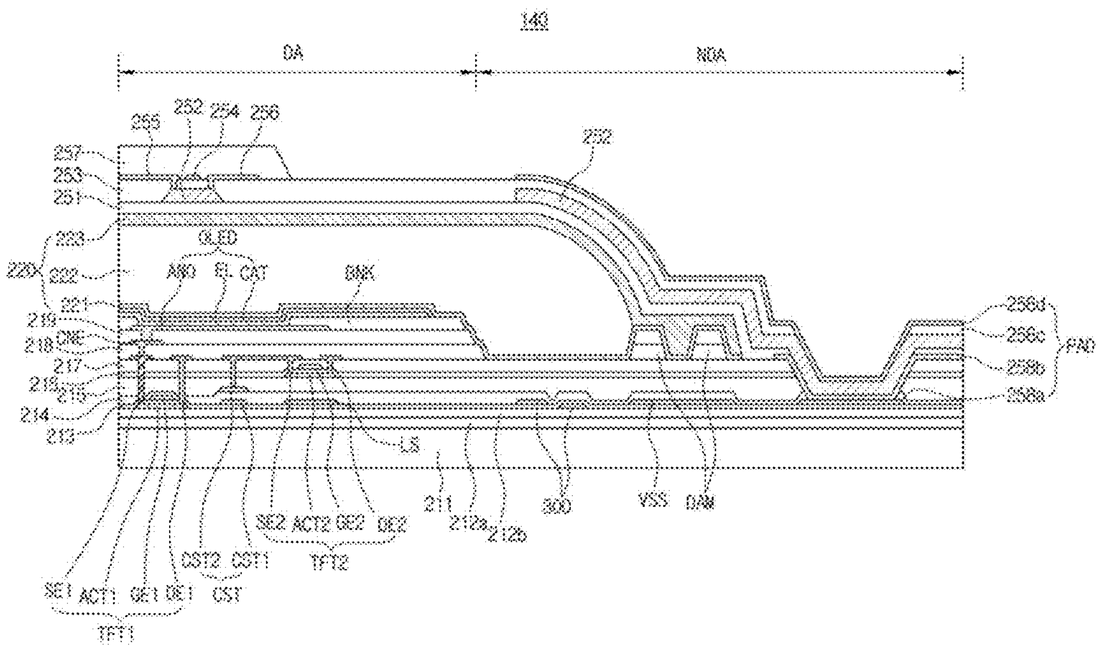
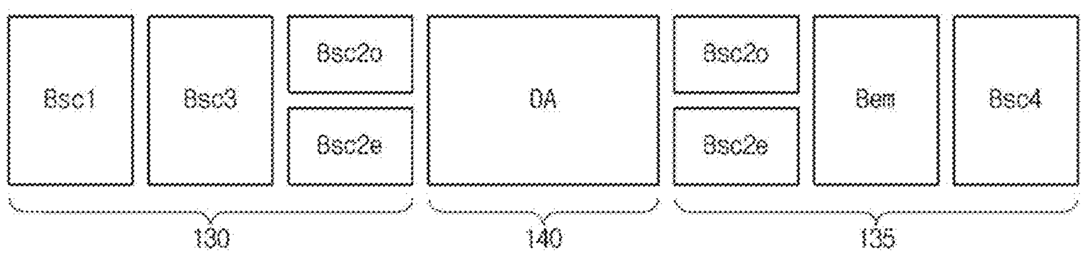


FIG. 2



140
DA
NDA
255
252
254
256
257
253
251
252
223
220
222
221
219
218
217
216
215
214
213
OLED
ANO
EL
CAT
BNK
CNE
256d
256c
256b
256a
PAD
LS
211
212a
212b
300
VSS
DAM
SE2
ACT2
GE2
DE2
TFT2
CST2
CST1
CST
SE1
ACT1
GE1
DE1
TFT1

FIG. 3



Bsc1
Bsc3
Bsc2o
Bsc2e
DA
Bsc2o
Bsc2e
Bem
Bsc4
130
140
135

FIG. 4

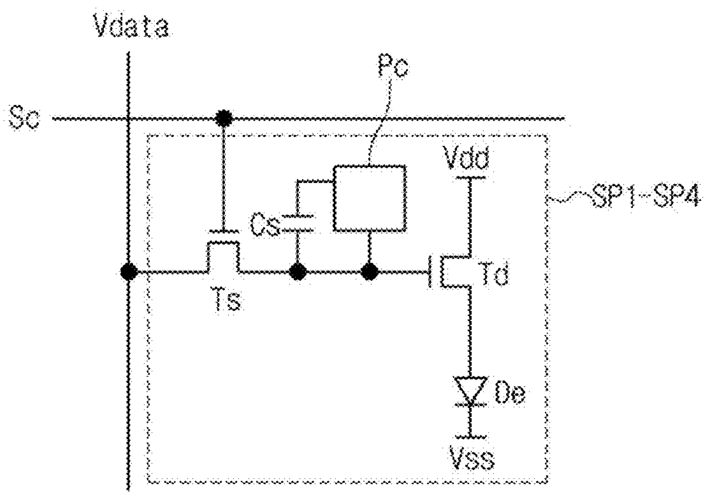


FIG. 5

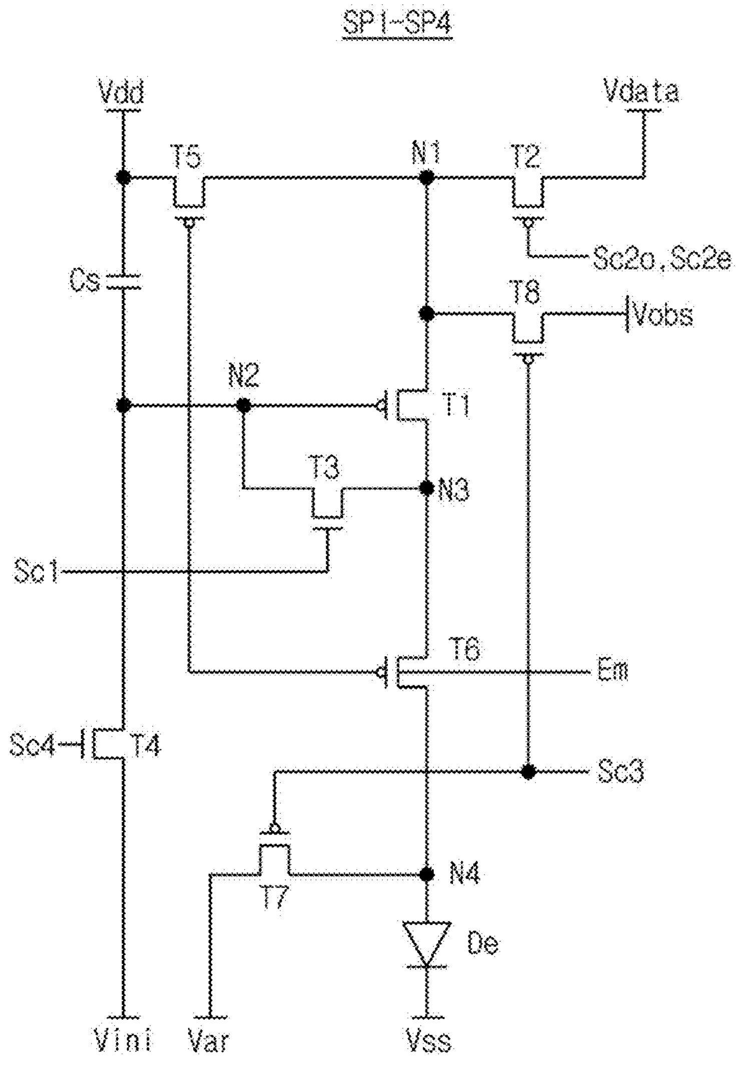
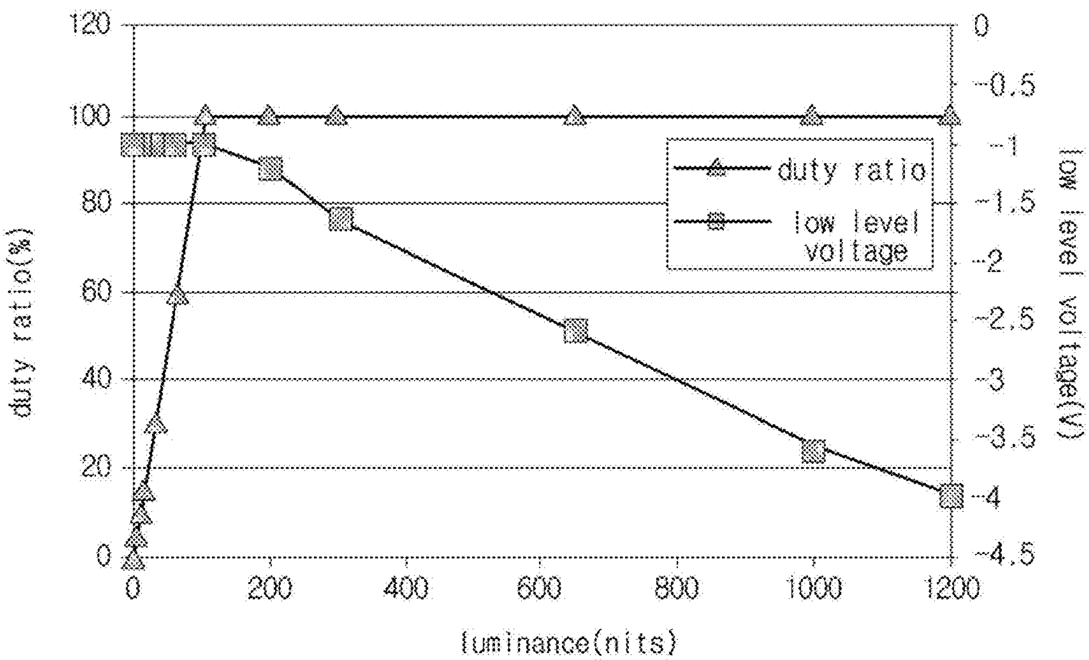
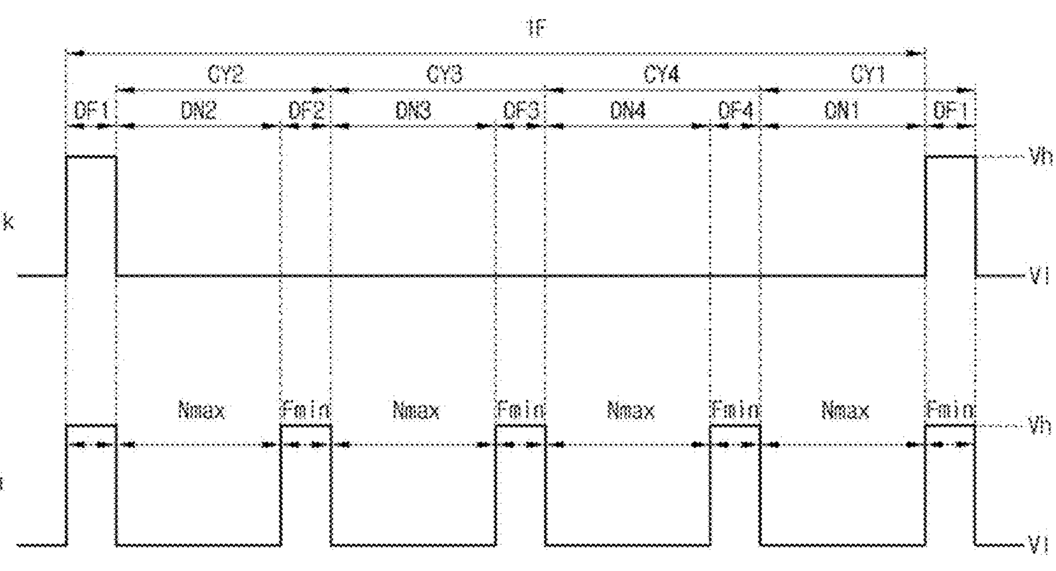

SP1~SP4
Vdd
T5
N1
T2
Vdata
Sc2o,Sc2e
Cs
T8
Vobs
N2
T1
T3
N3
Sc1
T6
Em
Sc4
T4
Sc3
N4
T7
De
Vini
Var
Vss

FIG. 6



120
100
80
60
40
20
0
0
-0.5
-1
-1.5
-2
-2.5
-3
-3.5
-4
-4.5
0
200
400
600
800
1000
1200
duty ratio(%)
low level voltage(V)
luminance(nits)
duty ratio
low level voltage

FIG. 7



1F
CY2
CY3
CY4
CY1
OF1
ON2
OF2
ON3
OF3
ON4
OF4
ON1
OF1
Vblk
Vh
Vl
Nmax
Fmin
Em
Vh
Vl

FIG. 8

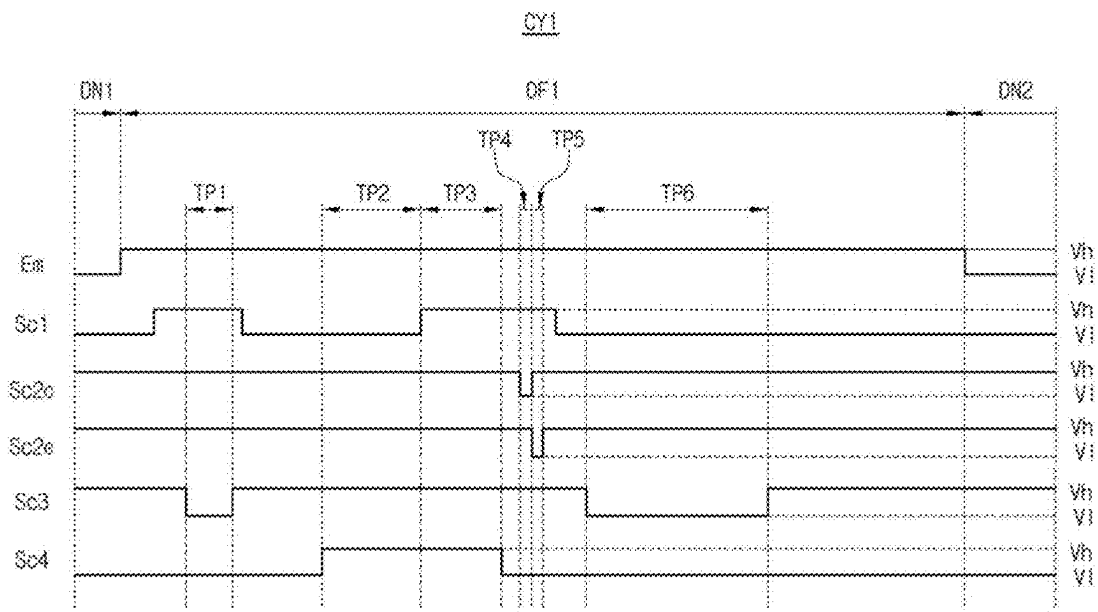

CY1
ON1
OF1
ON2
TP4
TP5
TP1
TP2
TP3
TP6
Em
Sc1
Sc2o
Sc2e
Sc3
Sc4
Vh
Vl

FIG. 9

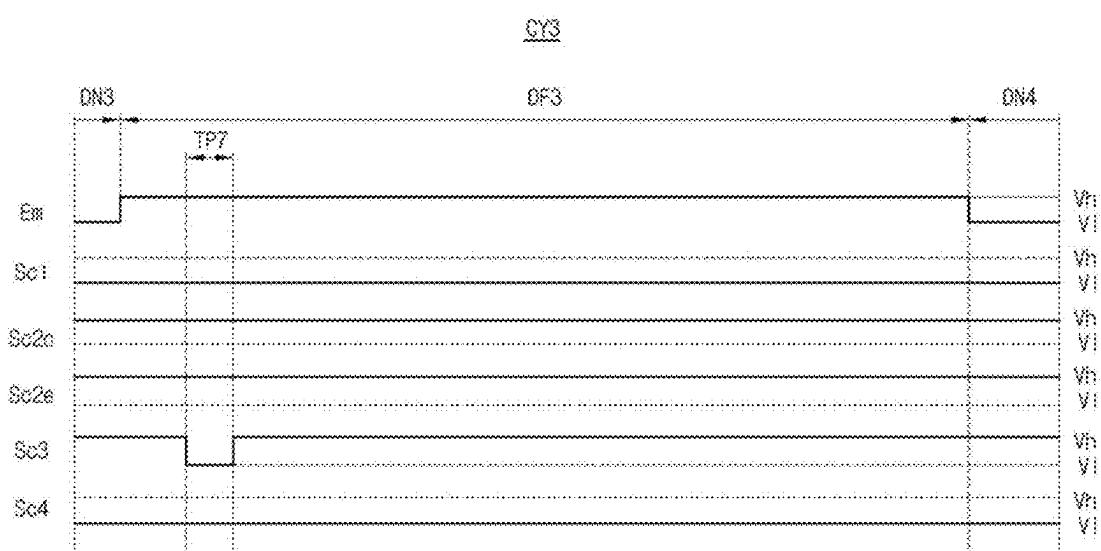

CY3
ON3
OF3
ON4
TP7
Em
Sc1
Sc2o
Sc2e
Sc3
Sc4
Vh
Vl

FIG. 10

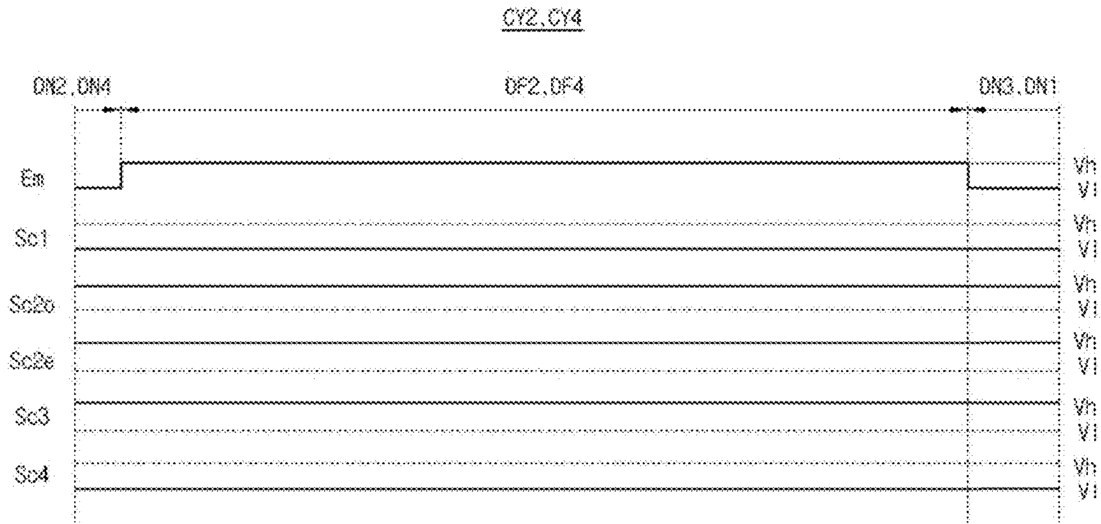

CY2, CY4
ON2, ON4
OF2, OF4
ON3, ON1
Em
Sc1
Sc2o
Sc2e
Sc3
Sc4
Vh
Vl

FIG. 11

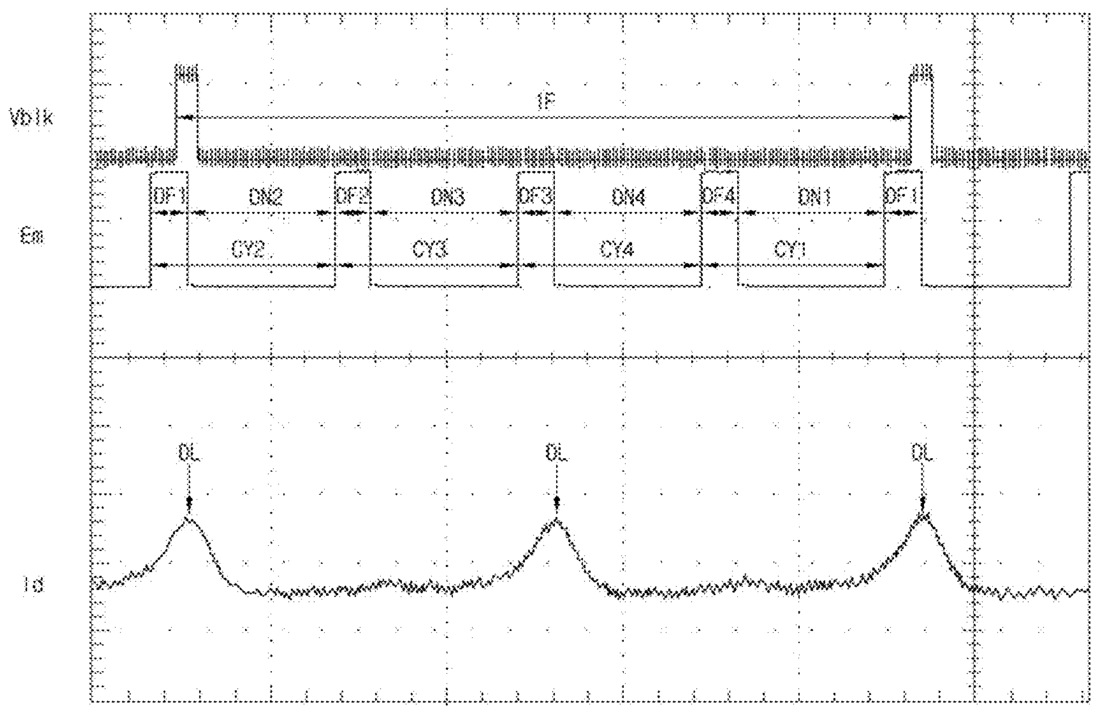
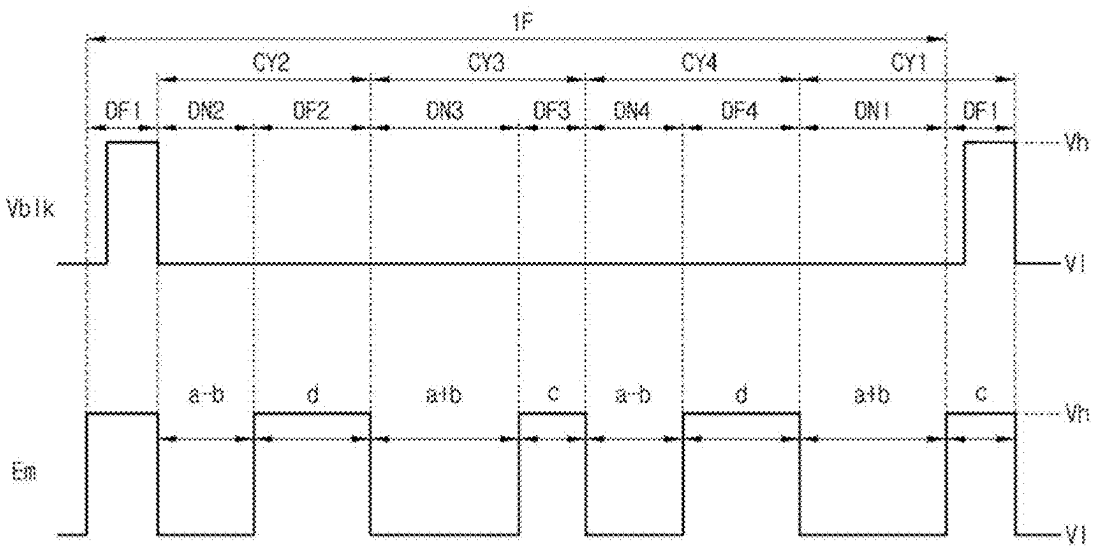

Vblk
1F
Em
OF1
ON2
OF2
ON3
OF3
ON4
OF4
ON1
CY2
CY3
CY4
CY1
OL
Id

FIG. 12


1F
CY2
CY3
CY4
CY1
OF1
ON2
OF2
ON3
OF3
ON4
OF4
ON1
OF1
Vblk
Vh
Vl
a-b
d
a+b
c
a-b
d
a+b
c
Em
Vh
Vl

DISPLAY DEVICE AND METHOD OF DRIVING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims the priority of Korean Patent Application No. 10-2024-0028278, filed on Feb. 27, 2024, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a display device.

Description of the Related Art

[0003] Recently, with the advent of an information-oriented society, the interest in information displays for processing and displaying a massive amount of information and the demand for portable information media have increased. As such, a display field has rapidly advanced. Thus, various light and thin flat panel display devices have been developed and highlighted.

[0004] Among the various flat panel display devices, an organic light emitting diode (OLED) display device is an emissive type device that does not include a backlight unit used in a non-emissive type device such as a liquid crystal display (LCD) device. As a result, the OLED display device has advantages in a viewing angle, a contrast ratio and a power consumption to be applied to various fields.

[0005] The OLED display device uses a pulse width modulation (PWM) when a luminance is controlled with the same voltage. In an emission state of a low gray level, since a voltage of an anode of a light emitting diode is changed due to a coupling, an abnormal phenomenon such as a luminance inversion, a gamma crush and a black rising occurs and a display quality of a low gray level image is deteriorated.

BRIEF SUMMARY

[0006] The present disclosure provides a display device that substantially obviates, among others, one or more of the problems due to limitations and disadvantages of the related art.

[0007] The disclosure provides a display device where a voltage variation of an anode of a light emitting diode due to a coupling is reduced by determining duty ratios of an emission signal of a plurality of cycles differently. For example, the present disclosure provides a display device where a voltage variation of an anode of a light emitting diode due to a coupling is reduced and a deterioration such as a gray level inversion, a gamma crush and a black rising is prevented by determining duty ratios of an emission signal of a plurality of cycles differently and a method of driving the display device.

[0008] Further, the present disclosure provides a display device where a voltage variation of an anode of a light emitting diode due to a coupling is reduced and a power consumption is reduced to obtain a low power driving by determining a duty ratio of an emission signal of a cycle having a reset period greater than a duty ratio of an emission signal of a cycle not having the reset period.

[0009] Additional features and improvements of the disclosure will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the disclosure. These and other advantages of the disclosure will be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

[0010] To achieve these and other features and improvement and in accordance with the purpose of the present disclosure, as embodied and broadly described herein, a display device includes: a timing controlling unit generating an image data, a data control signal and a gate control signal; a data driving unit generating a data signal using the image data and the data control signal; a gate driving unit generating a plurality of scan signals and an emission signal using the gate control signal; and a display panel displaying an image using the data signal, the plurality of scan signals and the emission signal, wherein the emission signal includes a plurality of cycles each having a duty-on period and an duty-off period during one frame, and wherein duty ratios of at least two of the plurality of cycles are different from each other.

[0011] In another aspect of the present disclosure, a method of driving a display device including a timing controlling unit, a data driving unit, a gate driving unit and a display panel includes: during a duty-on period of each of a plurality of cycles of one frame, emitting a light by a subpixel of the display panel; and during a duty-off period of each of the plurality of cycles, applying a data signal to the subpixel of the display panel, wherein duty ratios of at least two of the plurality of cycles are different from each other.

[0012] It is to be understood that both the foregoing general description and the following detailed description are explanatory and are intended to provide further explanation of the disclosure.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0013] The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification, illustrate embodiments of the disclosure and together with the description serve to explain the principles of the disclosure.

[0014] FIG. **1** is a view showing a display device according to an embodiment of the present disclosure;

[0015] FIG. **2** is a cross-sectional view showing a display panel of a display device according to an embodiment of the present disclosure;

[0016] FIG. **3** is a block diagram showing first and second gate driving units and a display panel of a display device according to an embodiment of the present disclosure;

[0017] FIG. **4** is a circuit diagram showing a subpixel of a display device according to an embodiment of the present disclosure;

[0018] FIG. **5** is a circuit diagram showing a subpixel of a 7T1C structure of a display device according to an embodiment of the present disclosure;

[0019] FIG. **6** is a view showing a luminance with respect to a duty ratio and a low level voltage of a display device according to an embodiment of the present disclosure;

[0020] FIG. **7** is a view showing a blank signal and an emission signal of one frame of a display device according to an embodiment of the present disclosure;

[0021] FIG. **8** is a view showing an emission signal and a scan signal of a first cycle of a display device according to an embodiment of the present disclosure;

[0022] FIG. **9** is a view showing an emission signal and a scan signal of a third cycle of a display device according to an embodiment of the present disclosure;

[0023] FIG. **10** is a view showing an emission signal and a scan signal of second and fourth cycles of a display device according to an embodiment of the present disclosure;

[0024] FIG. **11** is a view showing an emission signal and a luminance variation of first, second, third and fourth cycles of a display device according to an embodiment of the present disclosure; and

[0025] FIG. **12** is a view showing an on-duty and an off-duty of an emission signal of one frame of a display device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0026] Advantages and features of the present disclosure, and implementation methods thereof will be clarified through following example aspects described with reference to the accompanying drawings. The present disclosure may, however, be embodied in different forms and should not be construed as limited to the example aspects set forth herein. Rather, these example aspects are provided so that this disclosure may be sufficiently thorough and complete to assist those skilled in the art to fully understand the scope of the present disclosure. Further, the present disclosure includes those of the claims.

[0027] The shapes, sizes, ratios, angles, numbers, and the like, which are illustrated in the drawings to describe various example aspects of the present disclosure, are merely given by way of example. Therefore, the present disclosure is not limited to the illustrations in the drawings. Like reference numerals refer to like elements throughout the specification, unless otherwise specified.

[0028] In the following description, where the detailed description of the relevant known function or configuration may unnecessarily obscure a feature or aspect of the present disclosure, a detailed description of such known function or configuration may be omitted or a brief description may be provided.

[0029] Where the terms "comprise," "have," "include," and the like are used, one or more other elements may be added unless the term, such as "only," is used. An element described in the singular form is intended to include a plurality of elements, and vice versa, unless the context clearly indicates otherwise.

[0030] In construing an element, the element is to be construed as including an error or a tolerance range even where no explicit description of such an error or tolerance range is provided.

[0031] Where positional relationships are described, for example, where the positional relationship between two parts is described using "on," "over," "under," "above," "below," "beside," "next," or the like, one or more other parts may be located between the two parts unless a more limiting term, such as "immediate(ly)," "direct(ly)," or "close(ly)" is used. For example, where an element or layer is disposed "on" another element or layer, a third layer or element may be interposed therebetween.

[0032] Although the terms "first," "second," A, B, (a), (b), and the like may be used herein to refer to various elements, these elements should not be interpreted to be limited by these terms as they are not used to define a particular order or precedence. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure.

[0033] The term "at least one" should be understood to include all combinations of one or more of related elements. For example, the term of "at least one of first, second and third elements" may include all combinations of two or more of the first, second and third elements as well as the first, second or third element.

[0034] The term "display device" may include a display device in a narrow sense such as liquid crystal module (LCM), an organic light emitting diode (OLED) module and a quantum dot (QD) module including a display panel and a driving unit for driving the display panel. In addition, the term "display device" may include a complete product (or a final product) including the LCM, the OLED module and the QD module such as a notebook computer, a television, a computer monitor, an equipment display device including an automotive display apparatus or a shape other than a vehicle, and a set electronic apparatus or a set device (or a set apparatus) such as a mobile electronic apparatus of a smart phone or an electronic pad.

[0035] Accordingly, a display device of the present disclosure may include an applied product or a set device of a final user's device including the LCM, the OLED module and the QD module as well as a display device in a narrow sense such as the LCM, the OLED module and the QD module.

[0036] According to circumstances, the LCM, the OLED module and the QD module having a display panel and a driving unit may be expressed as "a display device", and an electronic apparatus of a complete product including the LCM, the OLED module and the QD module may be expressed as "a set device." For example, a display device in a narrow sense may include a display panel of a liquid crystal, an organic light emitting diode and a quantum dot and a source printed circuit board (PCB) of a control unit for driving the display panel, and a set device may further include a set PCB of a set control unit electrically connected to the source PCB for controlling the entire set device.

[0037] The display panel of the present disclosure may include all kinds of display panels such as a liquid crystal display panel, an organic light emitting diode display panel, a quantum dot display panel and an electroluminescent display panel. The display panel of the present disclosure is not limited to a specific display panel of a bezel bending having a flexible substrate for an organic light emitting diode display panel and a lower back plate supporter. A shape or a size of the display panel for the display device of the present disclosure is not limited thereto.

[0038] For example, when the display panel is an organic light emitting diode display panel, the display panel may include a plurality of gate lines, a plurality of data lines and a subpixel in a crossing region of the plurality of gate lines and the plurality of data lines. The display panel may include an array having a thin film transistor of an element for selectively applying a voltage to each subpixel, an emitting element layer on the array and an encapsulating substrate or an encapsulation part covering the emitting element layer. The encapsulation part may protect the thin film transistor and the emitting element layer from an external impact and may prevent or at least reduce penetration of a moisture or oxygen into the emitting element layer. In addition, the emitting element layer on the array may include an inorganic light emitting layer, for example, a nano-sized material layer or a quantum dot.

[0039] The thin film transistor of the present disclosure may include one of an oxide thin film transistor, an amorphous silicon thin film transistor, a low temperature polycrystalline silicon thin film transistor.

[0040] Features of various aspects of the present disclosure may be partially or entirely coupled to or combined with each other. They may be linked and operated technically in various ways as those skilled in the art may sufficiently understand. The aspects may be carried out independently of or in association with each other in various combinations.

[0041] Hereinafter, a display device according to various example aspects of the present disclosure where an influence on an oxide semiconductor layer of a thin film transistor of a driving element part is reduced by shielding a light emitted and transmitted from a subpixel and/or a light inputted from an exterior will be described in detail with reference to the accompanying drawings.

[0042] FIG. **1** is a view showing a display device according to an embodiment of the present disclosure. The display device may be an organic light emitting diode (OLED) display device.

[0043] In FIG. **1**, a display device **110** according to an embodiment of the present disclosure includes a timing controlling unit **120**, a data driving unit **125**, first and second gate driving units **130** and **135** and a display panel **140**.

[0044] The timing controlling unit **120** generates an image data, a data control signal and a gate control signal using an image signal and a plurality of timing signals including a data enable signal, a horizontal synchronization signal, a vertical synchronization signal and a clock signal transmitted from an external system such as a graphic card or a television system. The image data and the data control signal are transmitted to the data driving unit **125**, and the gate control signal is transmitted to the first and second gate driving units **130** and **135**.

[0045] The data driving unit **125** generates a data signal (a data voltage) Vdata (of FIGS. **4** and **5**) using the data control signal and the image data transmitted from the timing controlling unit **120** and transmits the data signal to a data line DL of the display panel **140**.

[0046] The first and second gate driving units **130** and **135** generate a gate signal (a gate voltage) Sc (of FIG. **4**), Sc**1**, Sc**2***o*, Sc**2***e*, Sc**3**, Sc**4** and Em (of FIG. **5**) using the gate control signal transmitted from the timing controlling unit **120** and applies the gate signal Sc, Sc**1**, Sc**2***o*, Sc**2***e*, Sc**3**, Sc**4** and Em to a gate line GL of the display panel **140**.

[0047] The first and second gate driving units **130** and **135** may have a gate in panel (GIP) type to be formed in a non-display area NDA of a substrate of the display panel **140** having the gate line GL, the data line DL and a pixel P.

[0048] Although the first and second gate driving units **130** and **135** are disposed in both side portions of the display panel **140** in the embodiment of FIG. **1**, one gate driving unit may be disposed in one side portion of the display panel **140** in another embodiment.

[0049] The display panel **140** includes a display area DA at a central portion thereof and a non-display area NDA surrounding the display area DA. The display panel **140** displays an image using the gate signal Sc, Sc**1**, Sc**2***o*, Sc**2***e*, Sc**3**, Sc**4** and Em and the data signal Vdata. For displaying an image, the display panel **140** includes a plurality of pixels P, a plurality of gate lines GL and a plurality of data lines DL in the display area DA.

[0050] Each of the plurality of pixels P includes first to fourth subpixels SP**1** to SP**4**, and the gate line GL and the data line DL cross each other to define the first to fourth subpixels SP**1** to SP**4**. Each of the first to fourth subpixels SP**1** to SP**4** is connected to the gate line GL and the data line DL. For example, the first to fourth subpixels SP**1** to SP**4** may correspond to red, green, blue and white colors, respectively.

[0051] When the display device **110** is an OLED display device, each of the first to fourth subpixels SP**1** to SP**4** may include a plurality of transistors such as a switching transistor Ts (of FIG. **4**), a driving transistor Td (of FIG. **4**) and a sensing transistor T**3** (of FIG. **5**), a storage capacitor Cs (of FIG. **4**) and a light emitting diode De (of FIG. **4**).

[0052] A structure of the display panel **140** and the subpixel SP of the display device **110** will be illustrated with reference to a drawing.

[0053] FIG. **2** is a cross-sectional view showing a display panel of a display device according to an embodiment of the present disclosure, FIG. **3** is a block diagram showing first and second gate driving units and a display panel of a display device according to an embodiment of the present disclosure, FIG. **4** is a circuit diagram showing a subpixel of a display device according to an embodiment of the present disclosure, and FIG. **5** is a circuit diagram showing a subpixel of a 7T1C structure of a display device according to an embodiment of the present disclosure.

[0054] In FIG. **2**, the display panel **140** of the display device **110** according to an embodiment of the present disclosure includes first and second thin film transistors TFT**1** and TFT**2** and a storage capacitor CST. The first and second thin film transistors TFT**1** and TFT**2** may include a polycrystalline semiconductor material or an oxide semiconductor material. For example, the first thin film transistor TFT**1** may include a polycrystalline semiconductor material, and the second thin film transistor TFT**2** may include an oxide semiconductor material.

[0055] The first thin film transistor TFT**1** is connected to a light emitting diode OLED, and the second thin film transistor TFT**2** is connected to the storage capacitor CST.

[0056] One subpixel SP includes the light emitting diode OLED and a pixel circuit supplying a driving current to the light emitting diode OLED. The pixel circuit is disposed on a substrate **211**, and the light emitting diode OLED is disposed in the pixel circuit. An encapsulating layer **220** is disposed on the light emitting diode OLED to protect the light emitting diode OLED.

[0057] The pixel circuit may include a driving thin film transistor, a switching thin film transistor and a storage capacitor. The light emitting diode OLED may include an anode, a cathode and an emitting layer between the anode and the cathode.

[0058] The driving thin film transistor and at least one switching thin film transistor use an oxide semiconductor material as an active layer. The thin film transistor using the oxide semiconductor material as an active layer has an excellent blocking effect for a leakage current and has a lower fabrication cost as compared with a thin film transistor using a polycrystalline semiconductor material as an active layer. As a result, to reduce a power consumption and a fabrication cost, the pixel circuit may include the driving thin film transistor and the at least one switching thin film transistor using the oxide semiconductor material.

[0059] For example, all of thin film transistors of the pixel circuit may be formed of the oxide semiconductor material, or a portion of the switching thin film transistors may be formed of the oxide semiconductor material.

[0060] The thin film transistor using the oxide semiconductor material has a relatively low reliability, while the thin film transistor using the polycrystalline semiconductor material has a relatively rapid operation speed and a relatively high reliability. As a result, the pixel circuit in an embodiment may include both of a switching thin film transistor using the oxide semiconductor material and a switching thin film transistor using the polycrystalline semiconductor material.

[0061] The substrate **211** may have a multiple layer of an organic layer and an inorganic layer alternately laminated. For example, the substrate **211** may include an organic layer of an organic insulating material such as polyimide and an inorganic layer of an inorganic insulating material such as silicon oxide (SiO_2) alternately laminated.

[0062] A lower buffer layer **212***a* is disposed on the substrate **211**. The lower buffer layer **212***a* may block a moisture penetrable from an exterior and may have a multiple layer including silicon oxide (SiO_2). An auxiliary buffer layer **212***b* for protecting elements from a moisture is disposed on the lower buffer layer **212***a*.

[0063] The first thin film transistor TFT**1** is disposed on the substrate **211**. The first thin film transistor TFT**1** may use a polycrystalline semiconductor material as an active layer. The first thin film transistor TFT**1** includes a first active layer ACT**1** having a channel where an electron or a hole moves, a first gate electrode GE**1**, a first source electrode SE**1** and a first drain electrode DE**1**.

[0064] The first active layer ACT**1** includes a first channel region, a first source region at one side of the first channel region and a first drain region at the other side of the first channel region.

[0065] The first source region and the first drain region include an intrinsic polycrystalline semiconductor material doped with an impurity of III or V group such as boron (B) or phosphorous (P). The first channel region includes an intrinsic polycrystalline semiconductor material to provide a path where an electron or a hole moves.

[0066] The first thin film transistor TFT**1** includes a first gate electrode GE**1** overlapping the first channel region of the first active layer ACT**1**. A first gate insulating layer **213** is disposed between the first gate electrode GE**1** and the first active layer ACT**1**. The first gate insulating layer **213** may have a single layer or a multiple layer of an inorganic insulating material such as silicon oxide (SiO_2) and silicon nitride (SiNx).

[0067] The first thin film transistor TFT**1** has a top gate structure where the first gate electrode GE**1** is disposed on the first active layer ACT**1**. As a result, a first capacitor electrode CST**1** of the storage capacitor CST and a light shielding layer LS of the second thin film transistor TFT**2** may have the same material as the first gate electrode GE**1**. A fabrication process may be simplified by forming the first gate electrode GE**1**, the first capacitor electrode CST**1** and the light shielding layer LS through one mask process.

[0068] The first gate electrode GE**1** may include a metallic material. For example, the first gate electrode GE**1** may have a single layer or a multiple layer of one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd), copper (Cu) and an alloy thereof.

[0069] A first interlayer insulating layer **214** is disposed on the first gate electrode GE**1**. For example, the first interlayer insulating layer **214** may include an inorganic insulating material such as silicon oxide (SiO_2) and silicon nitride (SiNx).

[0070] The display panel **140** may further include an upper buffer layer **215**, a second gate insulating layer **216** and a second interlayer insulating layer **217** sequentially disposed on the first interlayer insulating layer **214**. The first thin film transistor TFT**1** may include a first source electrode SE**1** and a first drain electrode DEI on the second interlayer insulating layer **217**, and the first source electrode SE**1** and the first drain electrode DEI may be connected to the first source region and the first drain region, respectively.

[0071] For example, the first source electrode SE**1** and the first drain electrode DEI may have a single layer or a multiple layer of one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd), copper (Cu) and an alloy thereof.

[0072] The upper buffer layer **215** separates a second active layer ACT**2** of an oxide semiconductor material of the second thin film transistor TFT**2** from the first active layer ACT**1** of a polycrystalline semiconductor material and provides a base for the second active layer ACT**2**.

[0073] The second gate insulating layer **216** covers the second active layer ACT**2** of the second thin film transistor TFT**2**. Since the second gate insulating layer **216** is disposed on the second active layer ACT**2** of an oxide semiconductor material, the second gate insulating layer **216** includes an inorganic insulating material. For example, the second gate insulating layer **216** may include silicon oxide (SiO_2) and silicon nitride (SiNx).

[0074] A second gate electrode GE**2** includes a metallic material. For example, the second gate electrode GE**2** may have a single layer or a multiple layer of one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd), copper (Cu) and an alloy thereof.

[0075] The second thin film transistor TFT**2** is disposed on the upper buffer layer **215** and includes the second active layer ACT**2** of an oxide semiconductor material, the second gate electrode GE**2** on the second gate insulating layer **216**, a second source electrode SE**2** and a second drain electrode DE**2** on the second interlayer insulating layer **217**.

[0076] The second active layer ACT**2** includes a second channel region, a second source region and a second drain region. The second channel region includes an intrinsic oxide semiconductor material which is not doped with an impurity, and the second source electrode and the second drain electrode are doped with an impurity to be conductorized.

[0077] The second thin film transistor TFT**2** is disposed above the upper buffer layer **215** and further includes a light shielding layer LS overlapping the second active layer ACT**2**. The light shielding layer LS blocks a light incident to the second active layer ACT**2** to obtain a reliability of the second thin film transistor TFT**2**. The light shielding layer LS may include the same material as the first gate electrode GE**1** and may be disposed on a top surface of the first gate insulating layer **213**. The light shielding layer LS may be electrically connected to the second gate electrode GE**2** to constitute a dual gate structure.

[0078] A fabrication process may be simplified by forming the second source electrode SE**2** and the second drain electrode DE**2** on the second interlayer insulating layer **217** simultaneously with the first source electrode SE**1** and the first drain electrode DE**1** through one mask process.

[0079] A second capacitor electrode CST**2** is disposed on the first interlayer insulating layer **214**. The second capacitor electrode CST**2** overlaps the first capacitor electrode CST**1** to constitute a storage capacitor CST. For example, the second capacitor electrode CST**2** may have a single layer or a multiple layer of one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd), copper (Cu) and an alloy thereof.

[0080] The storage capacitor CST stores the data signal supplied through the data line DL and supplies the data signal to the light emitting diode OLED. The storage capacitor CST includes two electrodes corresponding to each other and a dielectric layer between the two electrodes. A first interlayer insulating layer **214** is disposed between the first capacitor electrode CST**1** and the second capacitor electrode CST**2**.

[0081] One of the first and second capacitor electrodes CST**1** and CST**2** of the storage capacitor CST may be electrically connected to one of the second source electrode SE**2** and the second drain electrode DE**2** of the second thin film transistor TFT**2**. In another embodiment, a connection of the storage capacitor CST may be changed according to the pixel circuit.

[0082] A first planarizing layer **218** and a second planarizing layer **219** are sequentially disposed on the pixel circuit for planarizing the pixel circuit. For example, the first planarizing layer **218** and the second planarizing layer **219** may include an organic insulating material such as polyimide and acrylic resin.

[0083] A light emitting diode OLED is disposed on the second planarizing layer **219**.

[0084] The light emitting diode OLED includes an anode ANO, a cathode CAT and an emitting layer EL between the anode ANO and the cathode CAT. When the pixel circuit uses a low level voltage Vss (of FIG. **4**) connected to the cathode CAT commonly, the anode ANO may be disposed in each subpixel as an individual electrode. When the pixel circuit uses a high level voltage connected to the anode ANO commonly, the cathode CAT may be disposed in each subpixel as an individual electrode.

[0085] The light emitting diode OLED is electrically connected to a driving element through a central electrode CNE on the first planarizing layer **218**. The anode ANO of the light emitting diode OLED and the first source electrode SE**1** of the first thin film transistor TFT**1** of the pixel circuit are connected to each other through the central electrode CNE.

[0086] The anode ANO is connected to the central electrode CNE through a contact hole in the second planarizing layer **219**. The central electrode CNE is connected to the first source electrode SE**1** through a contact hole in the first planarizing layer **218**.

[0087] The central electrode CNE connects the first source electrode SE**1** and the anode ANO. For example, the central electrode CNE may include a conductive material such as copper (Cu), silver (Ag), molybdenum (Mo) and titanium (Ti).

[0088] The anode ANO may have a multiple layer including a transparent conductive layer and an opaque conductive layer having an excellent reflectance. For example, the transparent conductive layer may include a material having a relatively high work function such as indium tin oxide (ITO) and indium zinc oxide (IZO). The opaque conductive layer may have a single layer or a multiple layer of one of aluminum (Al), silver (Ag), copper (Cu), lead (Pb), molybdenum (Mo), titanium (Ti) and an alloy thereof. The anode ANO may have a structure such that a transparent conductive layer, an opaque conductive layer and a transparent conductive layer are sequentially laminated or a structure such that a transparent conductive layer and an opaque conductive layer are sequentially laminated.

[0089] The emitting layer EL includes a hole relating layer, an organic emitting layer and an electron relating layer sequentially or reversely laminated.

[0090] A bank layer BNK may be referred to as a pixel defining layer exposing the anode ANO of each subpixel SP**1** to SP**4**. The bank layer BNK may include an opaque material (e.g., a black material) to prevent a light interference between the adjacent subpixels SP**1** to SP**4**. The bank layer BNK may include a shielding material of at least one of a color pigment, an organic black and a carbon. A spacer may be disposed on the bank layer BNK.

[0091] The cathode CAT is disposed on a top surface and a side surface of the emitting layer EL to oppose the anode ANO. The cathode CAT may be disposed in the entire display area DA as one body. In a top emission type display device, the cathode CAT may include a transparent conductive material such as indium tin oxide (ITO) and indium zinc oxide (IZO).

[0092] An encapsulating layer **220** preventing permeation of a moisture may be disposed on the cathode CAT.

[0093] The encapsulating layer **220** may block permeation of a moisture or an oxygen of an exterior into the emitting layer EL. The encapsulating layer **220** may include at least one inorganic encapsulating layer and at least one organic encapsulating layer. The encapsulating layer **220** may exemplarily include a first encapsulating layer **221**, a second encapsulating layer **222** and a third encapsulating layer **223** in the display device **110**.

[0094] The first encapsulating layer **221** is disposed on the substrate **211** having the cathode CAT. The third encapsulating layer **223** is disposed on the substrate **211** having the second encapsulating layer **222** and wraps a top surface, a bottom surface and a side surface of the second encapsulating layer **222** with the first encapsulating layer **221**. The first encapsulating layer **221** and the third encapsulating layer **223** may minimize or prevent permeation of a moisture or an oxygen of an exterior into the emitting layer EL. For example, the first encapsulating layer **221** and the third encapsulating layer **223** may include an inorganic insulating material applicable to a low temperature deposition such as silicon nitride (SiNx), silicon oxide (SiOx), silicon oxynitride (SiON) and aluminum oxide (Al_2O_3). Deterioration of the emitting layer EL vulnerable to a relatively high temperature may be prevented by depositing the first encapsulating layer **221** and the third encapsulating layer **223** under a relatively low temperature.

[0095] The second encapsulating layer **222** may alleviate a stress between the layers of the display device **110** due to bending and may planarize a step difference of the layers of the display device **110**. For example, the second encapsulating layer **222** may be disposed on the substrate **211** having the first encapsulating layer **221** and may include a non-photosensitive organic insulating material such as acrylic resin, epoxy resin, phenolic resin, polyamide resin, polyimide resin, polyethylene and silicon oxycarbide (SiOC) or a photosensitive organic insulating material such as photoacryl. When the second encapsulating layer **222** is formed through an inkjet method, a dam DAM may be disposed to prevent diffusion of the liquid material for the second encapsulating layer **222** to an edge portion of the substrate **211**. The dam DAM may be disposed closer to the edge portion of the substrate **211** than the second encapsulating layer **222**. Due to the dam DAM, it is prevented that the second encapsulating layer **222** is diffused to a pad area of an outermost edge portion of the substrate **211** where a conductive pad is disposed.

[0096] Although the dam DAM is disposed to prevent diffusion of the second encapsulating layer **222**, a moisture may permeate the emitting layer through the exposed second encapsulating layer **222** when the second encapsulating layer **222** is formed higher than the dam DAM. As a result, the dam DAM may be formed to have a number of at least ten.

[0097] The dam DAM may be disposed on the second interlayer insulating layer **217** in the non-display area NDA.

[0098] The dam DAM may be formed simultaneously with the first planarizing layer **218** and the second planarizing layer **219**. For example, a lower layer of the dam DAM may be formed simultaneously with the first planarizing layer **218** and an upper layer of the dam DAM may be formed simultaneously with the second planarizing layer **219** such that the dam DAM has a double layered structure.

[0099] As a result, the dam DAM may have the same material as the first planarizing layer **218** and the second planarizing layer **219**.

[0100] The dam DAM may be disposed to overlap a low level voltage line VSS. For example, the low level voltage line VSS may be disposed under the dam DAM in the non-display area NDA.

[0101] The low level voltage line VSS and the first and second gate driving units **130** and **135** having a gate-in-panel (GIP) type are disposed to surround the display area DA of the display panel **140**, and the low level voltage line VSS may be disposed outside the first and second gate driving units **130** and **135**. Further, the low level voltage line VSS may be connected to the cathode CAT to supply a common voltage. Although the first and second gate driving units **130** and **135** are shown to have a simple structure in FIG. **1**, the first and second gate driving units **130** and **135** may include thin film transistors having the same structure as the thin film transistor of the display area DA.

[0102] For example, the low level voltage line VSS may have the same material as the first gate electrode GE**1** or the same material as the second capacitor electrode CST**2**, the first source electrode SE**1** and the first drain electrode DE**1**.

[0103] The low level voltage line VSS may supply a low level voltage Vss (of FIG. **4**) to the subpixel SP**1** to SP**4** in the display area DA.

[0104] A touch layer may be disposed on the encapsulating layer **220**. A touch buffer layer **251** of the touch layer may be disposed between a touch sensor metal and the cathode CAT of the light emitting diode OLED, and the touch sensor metal may include a touch connecting line **252** and **254** and a touch electrode **255** and **256**.

[0105] The touch buffer layer **251** may block permeation of a solution (a developing solution or an etching solution) used in a fabrication process of the touch sensor metal on the touch buffer layer **251** or a moisture of an exterior into the emitting layer EL including an organic material. As a result, the touch buffer layer **251** may prevent deterioration of the emitting layer EL susceptible to a solution or a moisture.

[0106] The touch buffer layer **251** includes an organic insulating material applicable to a low temperature lower than about 100° C. and having a dielectric constant of about 1 to about 3 to prevent deterioration of the emitting layer EL including an organic material vulnerable to a relatively high temperature. For example, the touch buffer layer **251** may include a material of an acrylic group, an epoxy group or a siloxane group. The touch buffer layer **251** of an organic insulating material having a planarization property may prevent deterioration of the encapsulating layer **220** due to a bending of the display device **110** and a breakdown of the touch sensor metal on the touch buffer layer **251**.

[0107] In a touch sensor structure based on a mutual capacitance, the touch electrodes **255** and **256** may be disposed on the touch buffer layer **251** and may alternate each other.

[0108] The touch connecting line **252** and **254** may connect the touch electrodes **255** and **256**. The touch connecting line **252** and **254** and the touch electrodes **255** and **256** may be disposed in different layers, and a touch insulating layer **253** may be disposed between the touch connecting line **252** and **254** and the touch electrodes **255** and **256**.

[0109] The touch connecting line **252** and **254** may be disposed to overlap the bank layer BNK to prevent reduction of an aperture ratio.

[0110] The touch electrodes **255** and **256** may be electrically connected to a touch driving circuit (not shown) through a portion of the touch connecting line **252** connected to a touch pad PAD passing through a top surface and a side surface of the encapsulating layer **220** and a top surface and a side surface of the dam DAM.

[0111] The portion of the touch connecting line **252** may receive a touch driving signal from the touch driving circuit and may transmit the touch driving signal to the touch electrode **255** and **256**. The portion of the touch connecting line **252** may transmit a touch sensing signal of the touch electrodes **255** and **256** to the touch driving circuit.

[0112] A touch protecting layer **257** may be disposed on the touch electrodes **255** and **256**. Although the touch protecting layer **257** is disposed on the touch electrodes **255** and **256** in an embodiment of FIG. **2**, the touch protecting layer **257** may extend a front or a rear of the dam DAM to be disposed on the touch connecting line **252**.

[0113] A color filter (not shown) may be disposed on the encapsulating layer **220**. The color filter may be disposed on the touch layer or may be disposed between the encapsulating layer **220** and the touch layer.

[0114] In FIG. **3**, the first gate driving unit **130** of the display device **110** includes a scan **1** block Bsc**1**, an odd scan**2** block Bsc**2***o*, an even scan**2** block Bsc**2***e* and a scan**3** block Bsc**3**, and the second gate driving unit **135** of the display device **110** includes an odd scan**2** block Bsc**2***o*, an even scan**2** block Bsc**2***e*, a scan**4** block Bsc**4** and an emission block Bem. The display area DA of the display panel **140** is disposed between the first and second gate driving units **130** and **135**.

[0115] In another embodiment, the disposition structure of the scan**1** block Bsc**1**, the odd scan**2** block Bsc**2***o*, the even scan**2** block Bsc**2***e*, the scan**3** block Bsc**3**, the scan**4** block Bsc**4** and the emission block Bem in the first and second gate driving units **130** and **135** may be variously changed.

[0116] For example, the scan**1** block Bsc**1** is disposed farther from the display panel **140** than the scan**3** block Bsc**3** and the scan**4** block Bsc**4** is disposed farther from the display panel **140** than the emission block Bem in an embodiment of FIG. **3**. In another embodiment, the scan**3** block Bsc**3** may be disposed farther from the display panel **140** than the scan**1** block Bsc**1** and the emission block Bem may be disposed farther from the display panel **140** than the scan**4** block Bsc**4**.

[0117] Each of the scan**1** block Bsc**1**, the odd scan**2** block Bsc**2***o*, the even scan**2** block Bsc**2***e* and the scan**3** block Bsc**3** of the first gate driving unit **130** and the odd scan**2** block Bsc**2***o*, the even scan**2** block Bsc**2***e*, the scan**4** block Bsc**4** and the emission block Bem of the second gate driving unit **135** may be one stage of a shift register, and the shift register may include a plurality of stages connected to each other in a cascade type.

[0118] In the first gate driving unit **130**, the scan**1** block Bsc**1**, the odd scan**2** block Bsc**2***o*, the even scan**2** block Bsc**2***e* and the scan**3** block Bsc**3** generate a scan**1** signal Sc**1** (of FIG. **5**), an odd scan**2** signal Sc**2***o* (of FIG. **5**), an even scan**2** signal Sc**2***e* (of FIG. **5**) and a scan**3** signal Sc**3** (of FIG. **5**), respectively.

[0119] In the second gate driving unit **135**, the odd scan**2** block Bsc**2***o*, the even scan**2** block Bsc**2***e*, the scan**4** block Bsc**4** and the emission block Bem generate the odd scan**2** signal Sc**2***o*, the even scan**2** signal Sc**2***e*, a scan**4** signal Sc**4** (of FIG. **5**) and an emission signal Em (of FIG. **5**), respectively.

[0120] The scan **1** signal Sc**1** of the scan**1** block Bsc**1** is supplied to a third transistor T**3** (of FIG. **5**) in each subpixel SP**1** to SP**4** of the display area DA through the gate line GL. The odd scan**2** signal Sc**2***o* of the odd scan**2** block Bsc**2***o* is supplied to a second transistor T**2** (of FIG. **5**) in each subpixel SP**1** to SP**4** of an odd horizontal pixel line of the display area DA through the gate line GL, and the even scan**2** signal Sc**2***e* of the even scan**2** block Bsc**2***e* is supplied to a second transistor T**2** in each subpixel SP**1** to SP**4** of an even horizontal pixel line of the display area DA through the gate line GL.

[0121] The scan**3** signal Sc**3** of the scan**3** block Bsc**3** is supplied to seventh and eighth transistors T**7** and T**8** (of FIG. **5**) in each subpixel SP**1** to SP**4** of the display area DA through the gate line GL. The scan**4** signal Sc**4** of the scan**4** block Bsc**4** is supplied to a fourth transistor T**4** (of FIG. **5**) in each subpixel SP**1** to SP**4** of the display area DA through the gate line GL, and the emission signal Em of the emission block Bem is supplied to fifth and sixth transistors T**5** and T**6** (of FIG. **5**) in each subpixel SP**1** to SP**4** of the display area DA through the gate line GL.

[0122] In another embodiment, the first and second gate driving units **130** and **135** may have a symmetric structure. For example, each of the first and second gate driving units **130** and **135** may include the scan **1** block Bsc**1**, the odd scan**2** block Bsc**2***o*, the even scan**2** block Bsc**2***e*, the scan**3** block Bsc**3**, the scan**4** block Bsc**4** and the emission block Bem.

[0123] In FIG. **4**, each of the first, second, third and fourth subpixels SP**1**, SP**2**, SP**3** and SP**4** of the display panel **140** of the display device **110** according to an embodiment of the present disclosure includes a switching transistor Ts, a driving transistor Td, a compensation part Pc, a storage capacitor Cs and a light emitting diode De. Active layers of the switching transistor Ts and the driving transistor Td may be formed of a semiconductor material, such as an oxide semiconductor material, amorphous semiconductor material, polycrystalline semiconductor material, or organic semiconductor material.

[0124] The oxide semiconductor material may have an excellent effect of preventing a leakage current and relatively inexpensive manufacturing cost. The oxide semiconductor may be formed of a metal oxide such as zinc (Zn), indium (In), gallium (Ga), tin (Sn), and titanium (Ti) or a combination of a metal such as zinc (Zn), indium (In), gallium (Ga), tin (Sn), or titanium (Ti) and its oxide. Specifically, the oxide semiconductor can include zinc oxide (ZnO), zinc-tin oxide (ZTO), zinc-indium oxide (ZIO), indium oxide (InO), titanium oxide (TiO), indium-gallium-zinc oxide (IGZO), indium-zinc-tin oxide (IZTO), indium zinc oxide (IZO), indium gallium tin oxide (IGTO), and indium gallium oxide (IGO).

[0125] The polycrystalline semiconductor material has a fast movement speed of carriers such as electrons and holes and thus has high mobility, and has low energy power consumption and superior reliability. The polycrystalline semiconductor may be formed of polycrystalline silicon (poly-Si).

[0126] The amorphous semiconductor material may be formed of amorphous silicon (a-Si).

[0127] For example, the switching transistor Ts and the driving transistor Td may be an oxide semiconductor thin film transistor, amorphous silicon thin film transistor or a low temperature polycrystalline silicon thin film transistor.

[0128] The switching transistor Ts is switched according to the scan signal Sc of the gate signal. A gate electrode of the switching transistor Ts is connected to the scan signal Sc, a source electrode of the switching transistor Ts is connected to a first capacitor electrode of the storage capacitor Cs and the compensation part Pc, and a drain electrode of the switching transistor Ts is connected to the data signal Vdata.

[0129] The driving transistor Td is switched according to a voltage of the first capacitor electrode of the storage capacitor Cs. A gate electrode of the driving transistor Td is connected to the first capacitor electrode of the storage capacitor Cs and the compensation part Pc, a source electrode of the driving transistor Td is connected to an anode of the light emitting diode De, and a drain electrode of the driving transistor Td is connected to the high level voltage Vdd.

[0130] The compensation part Pc is connected among the switching transistor Ts, the driving transistor Td and the storage capacitor Cs and compensates a variation of the threshold voltage Vth of the driving transistor Td.

[0131] The storage capacitor Cs stores the data signal Vdata. A first capacitor electrode of the storage capacitor Cs is connected to the source electrode of the switching transistor Ts and the compensation part Pc, and a second capacitor electrode of the storage capacitor Cs is connected to the compensation part Pc.

[0132] The light emitting diode De is connected between the driving transistor Td and the low level voltage Vss and emits a light of a luminance proportional to a current of the driving transistor Td. An anode of the light emitting diode De is connected to the source electrode of the driving transistor Td, and a cathode of the light emitting diode De is connected to the low level voltage Vss.

[0133] The data signal Vdata is supplied from the data driving unit **125** to each subpixel SP**1** to SP**4** of the display panel **140**, and the scan signal Sc is supplied from the first and second gate driving units **130** and **135** to each subpixel SP**1** to SP**4** of the display panel **140**.

[0134] In the pixel circuit of the present disclosure, various configurations of internal compensation circuits are possible. For example, a number of transistors TFTs in the pixel circuit of the present disclosure may be three or more, and a number of capacitors may be one or more. For example, each of the first to fourth subpixels SP**1** to SP**4** may have one of a 3T1C structure including three transistors (3T) and one capacitor (1C), a 6T1C structure including six transistors and one capacitor, a 7T1C structure including seven transistors and one capacitor and a 8T1C structure including eight transistors and one capacitor.

[0135] In FIG. **5**, each of the first to fourth subpixels SP**1** to SP**4** of the display panel **140** of the display device **110** according to an embodiment of the present disclosure includes first to eighth transistors T**1** to T**8**, a storage capacitor Cs and a light emitting diode De. At least one of the first to eighth transistors T**1** to T**8** may be an oxide semiconductor thin film transistor, and the others of the first to eighth transistors T**1** to T**8** may be low temperature polycrystalline silicon thin film transistor.

[0136] For example, the first, second, fifth, sixth, seventh and eighth transistors T**1**, T**2**, T**5**, T**6**, T**7** and T**8** may be a positive (P) type low temperature polycrystalline silicon thin film transistor, and the third and fourth transistors T**3** and T**4** may be a negative (N) type oxide semiconductor thin film transistor.

[0137] The first transistor T**1** as a driving transistor is switched according to a voltage of the first capacitor electrode of the storage capacitor Cs. A gate electrode of the first transistor T**1** is connected to a second node N**2**, a source electrode of the first transistor T**1** is connected to a first node N**1**, and a drain electrode of the first transistor T**1** is connected to a third node N**3**.

[0138] The second transistor T**2** as a switching transistor is switched according to an odd scan**2** signal Sc**2***o* or an even scan**2** signal Sc**2***e*. A gate electrode of the second transistor T**2** is connected to the odd scan**2** signal Sc**2***o* or the even scan**2** signal Sc**2***e*, a source electrode of the second transistor T**2** is connected to the first node N**1**, and a drain electrode of the second transistor T**2** is connected to the data signal Vdata.

[0139] The third transistor T**3** as a sensing transistor is switched according to a scan**1** signal Sc**1**. A gate electrode of the third transistor T**3** is connected to the scan **1** signal Sc**1**, a source electrode of the third transistor T**3** is connected to the third node N**3**, and a drain electrode of the third transistor T**3** is connected to the second node N**2**.

[0140] The fourth transistor T**4** as an initializing transistor is switched according to a scan**4** signal Sc**4**. A gate electrode of the fourth transistor T**4** is connected to the scan**4** signal Sc**4**, a source electrode of the fourth transistor T**4** is connected to an initial voltage Vini, and a drain electrode of the fourth transistor T**4** is connected to the second node N**2**.

[0141] The fifth transistor T**5** as an emitting transistor is switched according to an emission signal Em. A gate electrode of the fifth transistor T**5** is connected to the emission signal Em, a source electrode of the fifth transistor T**5** is connected to a high level voltage Vdd and the second capacitor electrode of the storage capacitor Cs, and a drain electrode of the fifth transistor T**5** is connected to the first node N**1**.

[0142] The sixth transistor T**6** as an emitting transistor is switched according to the emission signal Em. A gate electrode of the sixth transistor T**6** is connected to the emission signal Em, a source electrode of the sixth transistor T**6** is connected to the third node N**3**, and a drain electrode of the sixth transistor T**6** is connected to a fourth node N**4**.

[0143] The seventh transistor T**7** as a reset transistor is switched according to a scan**3** signal Sc**3**. A gate electrode of the seventh transistor T**7** is connected to the scan**3** signal Sc**3**, a source electrode of the seventh transistor T**7** is connected to the fourth node N**4**, and a drain electrode of the seventh transistor T**7** is connected to an anode reset signal (an anode reset voltage) Var.

[0144] The eighth transistor T**8** as a reset transistor is switched according to a scan**3** signal Sc**3**. A gate electrode of the eighth transistor T**8** is connected to the scan**3** signal Sc**3**, a source electrode of the eighth transistor T**8** is connected to the first node N**1**, and a drain electrode of the eighth transistor T**8** is connected to a stress signal (a stress voltage) Vobs.

[0145] The storage capacitor Cs stores the data signal Vdata and the threshold voltage Vth. A first capacitor electrode of the storage capacitor Cs is connected to the second node N**2**, and a second capacitor electrode of the storage capacitor Cs is connected to the high level voltage Vdd and the source electrode of the fifth transistor T**5**.

[0146] The light emitting diode De is connected between the sixth and seventh transistors T**6** and T**7** and the low level voltage Vss to emit a light of a luminance proportional to a current of the first transistor T**1**. An anode of the light emitting diode De is connected to the fourth node N**4**, and a cathode of the light emitting diode De is connected to the low level voltage Vss.

[0147] The source electrode of the first transistor T**1**, the source electrode of the second transistor T**2**, the drain electrode of the fifth transistor T**5** and the source electrode of the eighth transistor T**8** constitute the first node N**1**, and the gate electrode of the first transistor T**1**, the drain electrode of the third transistor T**3**, the first capacitor electrode of the storage capacitor Cs and the drain electrode of the fourth transistor T**4** constitute the second node N**2**. The drain electrode of the first transistor T**1**, the source electrode of the third transistor T**3** and the source electrode of the sixth transistor T**6** constitute the third node N**3**, and the drain electrode of the sixth transistor T**6**, the source electrode of the seventh transistor T**7** and the anode of the light emitting diode De constitute the fourth node N**4**.

[0148] In the display device **110**, a luminance of a light emitted from the light emitting diode De is adjusted using a pulse width modulation and a variable low level voltage.

[0149] FIG. **6** is a view showing a luminance with respect to a duty ratio and a low level voltage of a display device according to an embodiment of the present disclosure.

[0150] In FIG. **6**, the light emitting diode De of the display device **110** according to an embodiment of the present disclosure emits a light having a luminance smaller than about 100 nits due to a pulse width modulation (PWM) under the same voltage and emits a light of a luminance equal to or greater than about 100 nits due to a variable low level voltage Vss.

[0151] For example, the luminance of the light emitted from the light emitting diode De may increase within a range smaller than about 100 nits by increasing the duty ratio within a range of about 0% to about 100%, and the luminance of the light emitted from the light emitting diode De may increase within a range equal to or greater than about 100 nits by decreasing the low level voltage Vss within a range of about −0.75V to about −4V.

[0152] The duty ratio may be defined as a percentage of an emission period with respect to an entire high level period of a square wave.

[0153] The display device **110** displays a plurality of luminances using a plurality of luminance bands having different maximum luminance value according to an input of a user, and the plurality of luminance bands may correspond to a plurality of display brightness values (DBVs).

[0154] The display device **110** adjusts an image luminance using the display brightness values DBVs corresponding to a user's input. For example, when the luminance band includes first to thirteenth bands, each of the first to thirteenth bands may correspond to the display brightness values DBVs of 2047 to 117 of 11 bits and luminances of about 2175 nits to about 4 nits.

[0155] The seventh band may correspond to the display brightness value DBV of 505 of 11 bits and the luminance of about 100 nits. In the first to sixth bands where the display brightness value DBV is greater than the luminance of about 100 nits, the luminance is changed by adjusting the low level voltage Vss with the duty ratio fixed. In the seventh to thirteenth bands where the display brightness value DBV is equal to or smaller than the luminance of about 100 nits, the luminance is changed by adjusting the duty ratio with the low level voltage Vss fixed.

[0156] In the display device **110** according to an embodiment of the present disclosure, a power consumption may be reduced and a natural luminance change can be obtained by adjusting a luminance in a relatively low luminance range due to a pulse width modulation and adjusting a luminance in a relatively high luminance range due to a variable low level voltage.

[0157] In the display device **110**, one frame may be classified into a plurality of cycles.

[0158] FIG. **7** is a view showing a blank signal and an emission signal of one frame of a display device according to an embodiment of the present disclosure, FIG. **8** is a view showing an emission signal and a scan signal of a first cycle of a display device according to an embodiment of the present disclosure, FIG. **9** is a view showing an emission signal and a scan signal of a third cycle of a display device according to an embodiment of the present disclosure, and FIG. **10** is a view showing an emission signal and a scan signal of second and fourth cycles of a display device according to an embodiment of the present disclosure.

[0159] In FIG. **7**, one frame **1**F of the display device **110** according to an embodiment of the present disclosure includes first to fourth cycles CY**1** to CY**4**, and the first to fourth cycles CY**1** to CY**4** include a duty-on period DN**1** to DN**4** and a duty-off period DF**1** to DF**4**.

[0160] One frame **1**F may be classified according to a blank signal Vblk. The first cycle CY**1** may include a first duty-on period DN**1** where the emission signal Em has a logic low voltage Vl and the light emitting diode De is turned on (emits a light) and a first duty-off period DF**1** where the emission signal Em has a logic high voltage Vh and the light emitting diode De is turned off (does not emit a light).

[0161] Similarly, the second cycle CY**2** after the first cycle CY**1** may include a second duty-on period DN**2** and a second duty-off period DF**2**, the third cycle CY**3** after the second cycle CY**2** may include a third duty-on period DN**3** and a third duty-off period DF**3**, and the fourth cycle CY**4** after the third cycle CY**3** may include a fourth duty-on period DN**4** and a fourth duty-off period DF**4**.

[0162] As a result, in the display device **110** according to an embodiment of the present disclosure, the light emitting diode De emits a light four times of the first to fourth duty-on periods DN**1** to DN**4** during one frame **1**F.

[0163] When each of the first to fourth duty-on periods DN**1** to DN**4** has a maximum on-duty Nmax, each of the first to fourth duty-off periods DF**1** to DF**4** may have a minimum off-duty Fmin.

[0164] When the light emitting diode De emits a light of a maximum luminance of a plurality of luminance bands (a luminance adjustment by the pulse width modulation (PWM) and the fixed low level voltage Vss) where the display brightness value (DBV) is equal to or smaller than about 100 nits, each of the first to fourth duty-on periods DN**1** to DN**4** of the emission signal Em may have the maximum on-duty Nmax, and each of the first to fourth duty-off periods DF**1** to DF**4** of the emission signal Em may have the minimum off-duty Fmin.

[0165] For example, the maximum luminance, the maximum on-duty Nmax and the minimum off-duty Fmin of the plurality of luminance bands where the display brightness value (DBV) is equal to or smaller than about 100 nits may be about 100 nits, about 89.1% and about 10.9%, respectively.

[0166] Although one frame **1**F includes the first to fourth cycles CY**1** to CY**4** in an embodiment of FIG. **7**, one frame **1**F may include first and second cycles CY**1** and CY**2** or first to eighth cycles CY**1** to CY**8** in another embodiment.

[0167] In FIG. **8**, during a first period TP**1** of the first duty-off period DF**1** of the first cycle CY**1** of the display device **110** according to an embodiment of the present disclosure, the emission signal Em, the scan**1** signal Sc**1**, the odd scan**2** signal Sc**2***o* and the even scan**2** signal Sc**2***e* have a logic high voltage Vh, and the scan**3** signal Sc**3** and the scan**4** signal Sc**4** have a logic low voltage Vl.

[0168] As a result, the first, third, seventh and eighth transistors T**1**, T**3**, T**7** and T**8** are turned on, and the second, fourth, fifth and sixth transistors T**2**, T**4**, T**5** and T**6** are turned off. Accordingly, the stress voltage Vobs is applied to the first, third and second nodes N**1**, N**3** and N**2** through the eighth, first and third transistors T**8**, T**1** and T**3**, and the anode reset voltage Var is applied to the fourth node **N4** through the seventh transistor T**7**.

[0169] During the first period TP**1**, the first to fourth nodes **N1** to **N4** are reset, and a hysteresis phenomenon due to the previous frame is prevented.

[0170] During a second period TP**2**, the emission signal Em, the odd scan**2** signal Sc**2***o*, the even scan**2** signal Sc**2***e*, the scan**3** signal Sc**3** and the scan**4** signal Sc**4** have the logic high voltage Vh, and the scan**1** signal Sc**1** has the logic low voltage Vl.

[0171] As a result, the fourth transistor T**4** is turned on, and the first, second, third, fifth, sixth, seventh and eighth transistors T**1**, T**2**, T**3**, T**5**, T**6**, T**7** and T**8** are turned off. Accordingly, the initial voltage Vini is applied to the second node **N2** through the fourth transistor T**4**.

[0172] During the second period TP**2**, the second node **N2** is initialized to the initial voltage Vini.

[0173] During a third period TP**3**, the emission signal Em, the scan**1** signal Sc**1**, the odd scan**2** signal Sc**2***o*, the even scan**2** signal Sc**2***e*, the scan**3** signal Sc**3** and the scan**4** signal Sc**4** have the logic high voltage Vh.

[0174] As a result, the first, third and fourth transistors T**1**, T**3** and T**4** are turned on, and the second, fifth, sixth, seventh and eighth transistors T**2**, T**5**, T**6**, T**7** and T**8** are turned off. Accordingly, the initial voltage Vini is applied to the second, third and first nodes **N2**, **N3** and **N1** through the fourth, third and first transistors T**4**, T**3** and T**1**.

[0175] During the third period TP**3**, the second, third and first nodes **N2**, **N3** and **N1** are initialized to the initial voltage Vini.

[0176] During a fourth period TP**4**, the emission signal Em, the scan**1** signal Sc**1**, the even scan**2** signal Sc**2***e* and the scan**3** signal Sc**3** have the logic high voltage Vh, and the odd scan**2** signal Sc**2***o* and the scan**4** signal Sc**4** have the logic low voltage Vl.

[0177] As a result, the first, second and third transistors T**1**, T**2** and T**3** are turned on, and the fourth, fifth, sixth, seventh and eighth transistors T**4**, T**5**, T**6**, T**7** and T**8** are turned off. Accordingly, the data signal Vdata is applied to the second node **N2** through the second, first and third transistors T**2**, T**1** and T**3** of an odd horizontal pixel line.

[0178] During the fourth period TP**4**, the data signal Vdata is applied to the second node **N2** of the odd horizontal pixel line, and a sum (Vdata+Vth) of the data voltage Vdata and a threshold voltage Vth of the first transistor T**1** is applied to the gate electrode of the first transistor T**1** to be stored in the storage capacitor Cs.

[0179] During a fifth period TP**5**, the emission signal Em, the scan**1** signal Sc**1**, the odd scan**2** signal Sc**2***o* and the scan**3** signal Sc**3** have the logic high voltage Vh, and the even scan**2** signal Sc**2***e* and the scan**4** signal Sc**4** have the logic low voltage Vl.

[0180] As a result, the first, second and third transistors T**1**, T**2** and T**3** are turned on, and the fourth, fifth, sixth, seventh and eighth transistors T**4**, T**5**, T**6**, T**7** and T**8** are turned off. Accordingly, the data signal Vdata of an even horizontal pixel line is applied to the second node **N2** through the second, first and third transistors T**2**, T**1** and T**3**.

[0181] During the fifth period TP**5**, the data signal Vdata is applied to the second node of the even horizontal pixel line, and a sum (Vdata+Vth) of the data voltage Vdata and the threshold voltage Vth of the first transistor T**1** is applied to the gate electrode of the first transistor T**1** to be stored in the storage capacitor Cs.

[0182] During a sixth period TP**6**, the emission signal Em, the odd scan**2** signal Sc**2***o* and the even scan**2** signal Sc**2***e* have the logic high voltage Vh, and the scan**1** signal Sc**1**, the scan**3** signal Sc**3** and the scan**4** signal Sc**4** have the logic low voltage Vl.

[0183] As a result, the first, seventh and eighth transistors T**1**, T**7** and T**8** are turned on, and the second, third, fourth, fifth and sixth transistors T**2**, T**3**, T**4**, T**5** and T**6** are turned off. Accordingly, the stress voltage Vobs is applied to the first and third nodes **N1** and **N3** through the eighth and first transistors T**8** and T**1**, and the anode reset voltage Var is applied to the fourth node **N4** through the seventh transistor T**7**.

[0184] During the sixth period TP**6**, the first, third and fourth nodes **N1**, **N3** and **N4** are reset, and a hysteresis phenomenon due to the previous frame is prevented.

[0185] In FIG. **9**, during a seventh period TP**7** of the third duty-off period DF**3** of the third cycle CY**3** of the display device **110** according to an embodiment of the present disclosure, the emission signal Em, the odd scan**2** signal Sc**2***o* and the even scan**2** signal Sc**2***e* have the logic high voltage Vh, and the scan**1** signal Sc**1**, the scan**3** signal Sc**3** and the scan**4** signal Sc**4** have the logic low voltage Vl.

[0186] As a result, the seventh and eighth transistors T**7** and T**8** are turned on, and the first, second, third, fourth, fifth and sixth transistors T**1**, T**2**, T**3**, T**4**, T**5** and T**6** are turned off. Accordingly, the stress voltage Vobs is applied to the first node **N1** through the eighth transistor T**8**, and the anode reset voltage Var is applied to the fourth node **N4** through the seventh transistor T**7**.

[0187] During the seventh period TP**7**, the first and fourth nodes **N1** and **N4** are reset, and a hysteresis phenomenon due to the previous frame is prevented.

[0188] In FIG. **10**, during the second and fourth duty-off periods DF**2** and DF**4** of the second and fourth cycles CY**2** and CY**4** of the display device **110** according to an embodiment of the present disclosure, the emission signal Em, the odd scan**2** signal Sc**2***o*, the even scan**2** signal Sc**2***e* and the scan**3** signal Sc**3** have the logic high voltage Vh, and the scan**1** signal Sc**1** and the scan**4** signal Sc**4** have the logic low voltage Vl.

[0189] As a result, the first, second, third, fourth, fifth, sixth, seventh and eighth transistors T**1**, T**2**, T**3**, T**4**, T**5**, T**6**, T**7** and T**8** are turned off. Accordingly, a voltage of the previous cycle is maintained.

[0190] During the second and fourth duty-off periods DF**2** and DF**4** of the second and fourth cycles CY**2** and CY**4**, the voltage of the first, second, third and fourth nodes **N1**, **N2**, **N3** and **N4** is kept as the voltage of the first and third cycles CY**1** and CY**3** of the previous cycle.

[0191] During the first, second, third and fourth duty-on periods DN**1**, DN**2**, DN**3** and DN**4**, the odd scan**2** signal Sc**2***o*, the even scan**2** signal Sc**2***e* and the scan**3** signal Sc**3** have the logic high voltage Vh, and the emission signal Em, the scan**1** signal Sc**1** and the scan**4** signal Sc**4** have the logic low voltage Vl.

[0192] As a result, the first, third, fifth and sixth transistors T**1**, T**3**, T**5** and T**6** are turned on, and the second, third, fourth, seventh and eighth transistors T**2**, T**3**, T**4**, T**7** and T**8** are turned off. Accordingly, the high level voltage Vdd is applied to the fourth node **N4** through the fifth, first and sixth transistors T**5**, T**1** and T**6**. Since the threshold voltage Vth is compensated, a current corresponding to the data signal Vdata flows through the first transistor T**1** turned on.

[0193] During the first, second, third and fourth duty-on periods DN**1**, DN**2**, DN**3** and DN**4**, the light emitting diode De emits a light corresponding to the data signal Vdata of the present frame.

[0194] During the first and third duty-off periods DF**1** and DF**3** of the first and third cycles CY**1** and CY**3** of the display device **110** according to an embodiment of the present disclosure, the scan**3** signal has the logic low voltage Vl, and the seventh transistor T**7** is turned on. As a result, the fourth node N**4** is reset due to the anode reset voltage Var. During the second and fourth duty-off periods DF**2** and DF**4** of the second and fourth cycles CY**2** and CY**4**, the scan**3** signal Sc**3** does not have the logic low voltage Vl, and the seventh transistor T**7** is not turned on. As a result, the fourth node N**4** is not reset due to the anode reset voltage Var.

[0195] Accordingly, after the voltage of the fourth node N**4** rapidly decreases to the anode reset voltage Var during the first and sixth periods TP**1** and TP**6** of the first duty-off period DF**1**, the voltage of the fourth node N**4** gradually increases due to application of the high level voltage Vdd during the second duty-on period DN**2**, and the voltage of the fourth node N**4** increases and then decreases due to a kickback according to rising and falling of the emission signal Em during the second duty-off period DF**2**.

[0196] Next, after the voltage of the fourth node N**4** gradually increases due to application of the high level voltage Vdd during the third duty-on period DN**3** and the voltage of the fourth node N**4** increases and then decreases due to the kickback according to rising and falling of the emission signal Em during the third duty-off period DF**3**, the voltage of the fourth node N**4** rapidly decreases to the anode reset voltage Var during the seventh period TP**7** of the third duty-off period DF**3**.

[0197] Next, the voltage of the fourth node N**4** gradually increases due to application of the high level voltage Vdd during the fourth duty-on period DN**4**, and the voltage of the fourth node N**4** increases and then decreases due to the kickback according to rising and falling of the emission signal Em during the fourth duty-off period DF**4**.

[0198] The voltage of the fourth node N**4** rapidly decreases and a peak luminance decreases due to reset by the anode reset voltage Var during the first and third duty-off periods DF**1** and DF**3** of the first and third cycles CY**1** and CY**3**. As a result, a luminance variation of a light emitted from the light emitting diode De according to a voltage variation of the fourth node N**4** of the first and third duty-off periods DF**1** and DF**3** of the first and third cycles CY**1** and CY**3** is greater than a luminance variation of a light emitted from the light emitting diode De according to a voltage variation of the fourth node N**4** of the second and fourth duty-off periods DF**2** and DF**4** of the second and fourth cycles CY**2** and CY**4** where the rapid decrease (reset) of the voltage of the fourth node N**4** does not occur.

[0199] FIG. **11** is a view showing an emission signal and a luminance variation of first, second, third and fourth cycles of a display device according to an embodiment of the present disclosure.

[0200] In FIG. **11**, one frame **1**F of the display device **110** according to an embodiment of the present disclosure includes first, second, third and fourth cycles CY**1**, CY**2**, CY**3** and CY**4**. The first cycle CY**1** includes the first duty-on period DN**1** and the first duty-off period DF**1**, the second cycle CY**2** includes the second duty-on period DN**2** and the second duty-off period DF**2**, the third cycle CY**3** includes the third duty-on period DN**3** and the third duty-off period DF**3**, the fourth cycle CY**4** includes the fourth duty-on period DN**4** and the fourth duty-off period DF**4**.

[0201] The luminance of the light emitted from the light emitting diode De may be changed according to the voltage variation of the fourth node N**4** due to the kickback in the first, second, third and fourth duty-off periods DF**1**, DF**2**, DF**3** and DF**4**. The first and third duty-off periods DF**1** DF**3** where decrease of the peak luminance due to the anode reset voltage Var occurs have a relatively great luminance variation DL, while the second and fourth duty-off periods DF**2** and DF**4** where decrease of the peak luminance due to the anode reset voltage Var does not occur have a relatively small luminance variation.

[0202] As widths of the first, second, third and fourth duty-off periods DF**1**, DF**2**, DF**3** and DF**4** increase according to decrease of the duty ratio, a variation period of the emission signal Em increases and a voltage variation amount of the fourth node N**4** due to the kickback increases.

[0203] In the display device **110** according to an embodiment of the present disclosure, the duty-off period where decrease of the peak luminance due to the anode reset voltage Var does not occur further increases as compared with the duty-off period where decrease of the peak luminance due to the anode reset voltage Var occurs.

[0204] FIG. **12** is a view showing an on-duty and an off-duty of an emission signal of one frame of a display device according to an embodiment of the present disclosure.

[0205] In FIG. **12**, one frame **1**F of the display device **110** according to an embodiment of the present disclosure includes the first, second, third and fourth cycles CY**1**, CY**2**, CY**3** and CY**4**, and the first, second, third and fourth cycles CY**1**, CY**2**, CY**3** and CY**4** includes the first, second, third and fourth duty-on periods DN**1**, DN**2**, DN**3** and DN**4** and the first, second, third and fourth duty-off periods DF**1**, DF**2**, DF**3** and DF**4**.

[0206] As a result, the light emitting diode De emits a light four times of the first, second, third and fourth duty-on periods DN**1**, DN**2**, DN**3** and DN**4** during one frame **1**F.

[0207] When the display device **110** is driven with a predetermined duty ratio, the widths of the first and third duty-off periods DF**1** and DF**3** where rapid decrease of the voltage of the fourth node N**4** due to the anode reset voltage Var occurs are determined smaller than the widths of the second and fourth duty-off periods DF**2** and DF**4** where rapid decrease of the voltage of the fourth node N**4** due to the anode reset voltage Var does not occur, and the widths of the first and third duty-on periods DN**1** and DN**3** corresponding to the first and third duty-off periods DF**1** and DF**3** are determined greater than the widths of the second and fourth duty-on periods DN**2** and DN**4** corresponding to the second and fourth duty-off periods DF**2** and DF**4**. As a result, the luminance variation due to the kickback according to the emission signal Em is reduced or minimized.

[0208] For example, when the display device **110** is driven with the duty ratio of a, each of the first and third duty-on periods DN**1** and DN**3** may be determined to have the on-duty of (a+b), and each of the second and fourth duty-on periods DN**2** and DN**4** may be determined to have the on-duty of (a−b) smaller than (a+b).

[0209] Each of the first and third duty-off periods DF**1** and DF**3** may have the duty ratio of c, and each of the second and fourth duty-off periods DF**2** and DF**4** may have the duty ratio of d greater than c. (a+b+c=100%, a−b+d=100%, c=100%−(a+b), d=100%−a+b=c+2b; a, b, c are a positive integer)

[0210] When the display device **110** is driven with the duty ratio of about 80%, each of the first and third duty-on periods DN**1** and DN**3** may be determined to have the on-duty of about 90%, and each of the second and fourth duty-on periods DN**2** and DN**4** may be determined to have the on-duty of about 70%. Each of the first and third duty-off periods DF**1** and DF**3** may be determined to have the off-duty of about 10%, and each of the second and fourth duty-off periods DF**2** and DF**4** may be determined to have the off-duty of about 30%.

[0211] Further, each of the on-duty of (a+b) and the on-duty of (a−b) may be smaller than the maximum on-duty Nmax, and each of the off-duty of c and the off-duty of d may be greater than the minimum off-duty Fmin.

[0212] In the display device **110** according to an embodiment of the present disclosure, the width of the duty-off period where decrease of the peak luminance occurs is determined to be smaller than the width of the duty-off period where decrease of the peak luminance does not occur, and the width of the duty-on period corresponding to the duty-off period where decrease of the peak luminance occurs is determined to be greater than the width of the duty-on period corresponding to the duty-off period where decrease of the peak luminance does not occur. Since the voltage variation of the anode of the light emitting diode due to the coupling of the emission signal is reduced or minimized, the deterioration such as the gray level inversion, the gamma crush and the black rising is prevented, and the power consumption is reduced to obtain the low power driving.

[0213] It will be apparent to those skilled in the art that various modifications and variation may be made in the present disclosure without departing from the scope of the disclosure. Thus, it is intended that the present disclosure cover the modifications and variations of this disclosure including those of the appended claims and their equivalents.

[0214] The various embodiments described above can be combined to provide further embodiments. Aspects of the embodiments can be modified, if necessary to employ concepts of the various embodiment to provide yet further embodiments.

[0215] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

1. A display device, comprising:
- a timing controlling unit configured to generate an image data, a data control signal and a gate control signal;
- a data driving unit configured to generate a data signal using the image data and the data control signal;
- a gate driving unit configured to generate a plurality of scan signals and an emission signal using the gate control signal; and
- a display panel configured to display an image using the data signal, the plurality of scan signals and the emission signal,
- wherein the emission signal includes a plurality of cycles each having a duty-on period and an duty-off period during one frame, and
- wherein duty ratios of at least two of the plurality of cycles are different from each other.

2. The display device of claim **1**, wherein a duty ratio of one of the plurality of cycles having a reset period is greater than a duty ratio of another of the plurality of cycles where the reset period is omitted.

3. The display device of claim **1**, wherein the plurality of cycles include first, second, third and fourth cycles,
- wherein the duty-off period of the first and third cycles includes a reset period,
- wherein the reset period is omitted in the duty-off period of the second and fourth cycles,
- wherein a width of the duty-off period of the first and third cycles is smaller than a width of the duty-off period of the second and fourth cycles, and
- wherein a width of the duty-on period of the first and third cycles is greater than a width of the duty-on period of the second and fourth cycles.

4. The display device of claim **3**, wherein the display panel is driven with a duty ratio of a,
- wherein the duty-on period of the first and third cycles has an on-duty of (a+b), and the duty-on period of the second and fourth cycles has an on-duty of (a−b), and
- wherein the duty-off period of the first and third cycles has an on-duty of c, and the duty-off period of the second and fourth cycles has an on-duty of (c+2b), a, b, c being a positive integer, and a+b+c=100%.

5. The display device of claim **1**, wherein the plurality of scan signals include a scan**1** signal, an odd scan**2** signal, an even scan**2** signal, a scan**3** signal and a scan**4** signal,
- wherein the display panel includes a plurality of subpixels, and
- wherein each of the plurality of subpixels comprises:
  - a storage capacitor connected to a high level voltage;
  - a first transistor configured to be switched according to a voltage of a first capacitor electrode of the storage capacitor;
  - a second transistor configured to be switched according to one of the odd scan**2** signal and the even scan**2** signal and connected to the data signal and the first transistor;
  - a third transistor configured to be switched according to the scan**1** signal and connected to the storage capacitor and the first transistor;
  - a fourth transistor configured to be switched according to the scan**4** signal and connected to the storage capacitor and an initial voltage;
  - a fifth transistor configured to be switched according to the emission signal and connected to the high level voltage and the first transistor;
  - a sixth transistor configured to be switched according to the emission signal and connected to the first transistor;
  - a seventh transistor configured to be switched according to the scan**3** signal and connected to an anode reset voltage and the sixth transistor;
  - an eighth transistor configured to be switched according to the scan**3** signal and connected to a stress voltage and the first transistor; and
  - a light emitting diode connected between the sixth transistor and a low level voltage.

6. The display device of claim 5, wherein at least one of the first to eighth transistors is an oxide semiconductor thin film transistor.

7. The display device of claim 5, wherein the plurality of cycles include first, second, third and fourth cycles,
wherein the duty-off period of the first and third cycles includes first, second, third, fourth, fifth and sixth periods,
wherein, during the first period, the first, third, seventh and eighth transistors are turned on, the second, fourth, fifth and sixth transistors are turned off, the stress voltage is applied to first, third and second nodes, and the anode reset voltage is applied to a fourth node,
wherein, during the second period, the fourth transistor is turned on, the first, second, third, fifth, sixth, seventh and eighth transistors are turned off, and the initial voltage is applied to the second node,
wherein, during the third period, the first, third and fourth transistors are turned on, the second, fifth, sixth, seventh and eighth transistors are turned off, and the initial voltage is applied to the second, third and first nodes,
wherein, during the fourth period, the first, second and third transistors are turned on, the fourth, fifth, sixth, seventh and eighth transistors are turned off, and the data signal is applied to the second nodes,
wherein, during the fifth period, the first, second and third transistors are turned on, the fourth, fifth, sixth, seventh and eighth transistors are turned off, and the data signal is applied to the second node, and
wherein, during the sixth period, the first, seventh and eighth transistors are turned on, the second, third, fourth, fifth and sixth transistors are turned off, the stress voltage is applied to the first and third nodes, and the anode reset voltage is applied to the fourth node.

8. The display device of claim 7, wherein the duty-off period of the second and fourth cycles includes seventh, eighth, ninth and tenth periods,
wherein, during the seventh period, the fourth transistor is turned on, the first, second, third, fifth, sixth, seventh and eighth transistors are turned off, and the initial voltage is applied to the second node,
wherein, during the eighth period, the first, third and fourth transistors are turned on, the second, fifth, sixth, seventh and eighth transistors are turned off, and the initial voltage is applied to the second, third and first nodes,
wherein, during the ninth period, the first, second and third transistors are turned on, the fourth, fifth, sixth, seventh and eighth transistors are turned off, and the data signal is applied to the second node, and
wherein, during the tenth period, the first, second and third transistors are turned on, the fourth, fifth, sixth, seventh and eighth transistors are turned off, and the data signal is applied to the second node.

9. The display device of claim 1, wherein the plurality of scan signals include a scan1 signal, an odd scan2 signal, an even scan2 signal, a scan3 signal and a scan4 signal,
wherein the gate driving unit includes first and second gate driving units disposed at both sides of the display panel,
wherein the first gate driving unit includes a scan1 block generating the scan1 signal, an odd scan2 block generating the odd scan2 signal, an even scan2 block generating the even scan2 signal and a scan3 block generating the scan3 signal, and
wherein the second gate driving unit includes an odd scan2 block generating the odd scan2 signal, an even scan2 block generating the even scan2 signal, a scan4 block generating the scan4 signal and an emission block generating the emission signal.

10. The display device of claim 9, wherein the scan1 block is disposed farther from the display panel than the scan3 block, or the scan3 block is disposed farther from the display panel than the scan1 block, and
wherein the scan4 block is disposed farther from the display panel than the emission block, or the emission block is disposed farther from the display panel than the scan4 block.

11. The display device of claim 1, wherein each of the plurality of cycles includes a duty-on period and a duty-off period, and
wherein the width of the duty-off period where decrease of the peak luminance occurs is smaller than the width of the duty-off period where decrease of the peak luminance does not occur.

12. A method of driving a display device including a timing controlling unit, a data driving unit, a gate driving unit and a display panel, comprising:
during a duty-on period of each of a plurality of cycles of one frame, emitting a light by a subpixel of the display panel; and
during a duty-off period of each of the plurality of cycles, applying a data signal to the subpixel of the display panel,
wherein duty ratios of at least two of the plurality of cycles are different from each other.

13. The method of claim 12, wherein a duty ratio of one of the plurality of cycles having a reset period is greater than a duty ratio of another one of the plurality of cycles that does not have a reset period.

14. The method of claim 12, wherein the plurality of cycles include first, second, third and fourth cycles,
wherein the duty-off period of the first and third cycles includes a reset period,
wherein the reset period is omitted in the duty-off period of the second and fourth cycles,
wherein a width of the duty-off period of the first and third cycles is smaller than a width of the duty-off period of the second and fourth cycles, and
wherein a width of the duty-on period of the first and third cycles is greater than a width of the duty-on period of the second and fourth cycles.

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